

Date: 4/22/2025

Project No.: 24-046

Project Name: IPA Church

Reviewed By: Rob DuBoice

Shop Drawing No. 230000-13 Energy Recovery Units



REVIEWED



REVISE AND RESUBMIT



REVIEWED AS NOTED



REJECTED

This review is only for general conformance with the design concept and the information given in the Construction Documents. Corrections of comments made on the shop drawings during this review do not relieve the contractor from compliance with the requirements of the plans and specifications or review of a specific item shall not include review of an assembly of which the item is a component. The contractor is responsible for: dimensions to be confirmed and correlated at the jobsite; information that pertains solely to the fabrication processes or to the means, methods, techniques, sequences and procedures of construction; coordination of the work with that of all other trades and performing all work in a safe and satisfactory manner.

By Rob DuBoice Date 4/22/2025
Design Management Group, Pittston Pa

-Coordinate installation of unit with all other trades – including (but not limited to) architectural, structural, plumbing, and electrical.

Core Concepts Contracting, Inc.
P.O. Box 11041
Hauppauge, NY 11788
(631) 326-8985



SUBMITTAL

Project: C24-001
India Pentecostal Assembly
310A South Oyster Bay Road
Syosset, New York 11791

Spec Section No: 230000
Submittal No: 230000-13
Revision No: 00
Sent Date: 4/3/2025

Spec Section Title: HVAC
Submittal Title: Energy Recovery Units
Submittal Type: Shop Drawing

Contractor:
Core Concepts Contracting, Inc.

Contractor's Stamp

CORE CONCEPTS CONTRACTING, INC.

PROJECT# C24-001

SUBMITTAL # 230000-13

CONTRACTOR Northern Wolves

LOGGED BY JP *DATE* 4/3/2025

Architect:
Kent Architectural Collaborative

Architect's Stamp

Engineer:
Design Management Group

Engineer's Stamp



Northern Wolves AC
55 9th St, 55-A2
Brooklyn, New York 11215
P: (347) 463-9248

Project: 2716 India Pentecostal Assembly
310 South Oyster Bay Rd
Syosset, New York 11791

Submittal #13.0 - Greenheck Energy Recovery Ventilation Unit

Revision	0	Submittal Manager	Christino Ayala (Northern Wolves AC)
Status	Open	Date Created	Apr 1, 2025
Issue Date	Apr 1, 2025	Spec Section	
Responsible Contractor	Northern Wolves AC	Received From	
Received Date		Submit By	
Final Due Date		Lead Time	
		Cost Code	15010 - SUBMITTALS
Location		Type	Product Manual
Submittal Package			
Approvers			
Ball in Court	Christino Ayala (Northern Wolves AC)		
Distribution			
Description	ERV		

Submittal Workflow

Name	Sent Date	Due Date	Returned Date	Response	Attachments
General Information Attachments					P281653R03 - India pentecostal assembly -RVE Submittal (1).pdf



Submittal

Date: 3/25/2025

Job Title INDIA PENTECOSTAL ASSEMBLY
Contractor Northern Wolves
Engineer DMG Droup
Drawings Dated 2/6/2025
Submittal Prepared By: Ryan Adams /Julio Enriquez (ADEsolutions@adehvac.com)

Greenheck Energy Recovery Ventilation Unit

Tags Provided: (ERU-1,2,3)

For Approval

Items Included

Field Installed Features	Factory Installed Features
Shipped Loose Smoke Detectors - Both Airstreams	Model RVE-85-52D-20I-J-A2 (ERU-1,2,3)
14" Roof Curbs	High Turndown - 12:1 Stainless Steel Modulating IG Furnace - Size 300 MBH
Energy Wheel Warranty - 5 Yrs Less Motor; Unit Warranty: 1-Year (Parts Only)	Packaged DX Cooling: 20.0 Tons; Compressor Type: Inverter Scroll
	Modulating Hot Gas Reheat; EC Condenser Fan: Lead
	Refrigerant Type: R-454B; UL/cUL60335
	Weatherhood: Downturned Hood; Expansion Board
	Frost Control: Modulating Wheel
	Outdoor & Exhaust Air Filters - MERV 8, 8-16x16x2
	Supply Filters - 2" Pleated MERV 13, 3-16x20x2, 6-20x20x2
	Outdoor & Return Air Dampers - Motorized Low Leakage w/ Damper End Switch
	Unit Disconnect - Mounted By Factory;
	Painted Exterior - Concrete Gray (RAL 7023)
	Microprocessor Controls
	Supply Fan VFD Control - CO2 - By Factory - Room/Space
	Exhaust Fan VFD Control - Space Static Pressure by Factory
	Economizer Mode - Temp./Enthalpy; OA RH Sensor
	Wheel Control - Modulating Wheel
	Space Temp and RH - 1 Temp and RH, 3 Additional Temp Only (avg)
	Network Protocol: BACNetMSTP; Remote Display w/150' Cord; Rotation Sensor
	Phase and Brown Out Protection; Branch Circuit Fusing
	120v NEMA 3R Outlet (Factory mounted and wired); Condensate Overflow

****Items not listed above will not be included****

Exceptions/Not Included

Start-Up	unit color as selected by architect
6" equipment pad	DDC Actuator

****Please Note Air Pattern, Voltage, & Access, and Confirm Prior to Release****

For Complete Product Listing, Please Visit www.adehvac.com

SUBMITTED MATERIAL WITHIN IS SUBJECT TO PRICE VALIDATION AFTER 4/25/2025

TERMS AND CONDITIONS OF SALE

The following Conditions of Sale take precedence over - and supercede - all conditions set forth in any purchase order issued by Purchaser.

1. If the price of any item is not set forth on the invoice, it is Purchaser's obligation to advise Seller herein, in writing, at the address set forth on the reverse hereof, of the specific incorrect item(s), within 48 hours of receipt of this invoice.

2. All materials furnished come ONLY WITH THE WARRANTY OF THEIR MANUFACTURER. THE SELLER SPECIFICALLY HEREBY CLEARLY STATES THAT NO WARRANTY OF ANY KIND, INCLUDING THE WARRANTIES OF MERCHANTABILITY AND FITNESS FOR USE IS PROVIDED BY THIS SELLER. Copies of the various manufacturers' warranties are available upon request.

3. MANUFACTURERS' WARRANTIES WILL ONLY APPLY TO THOSE MATERIALS (INCLUDING INSTALLATION) WHICH HAVE BEEN PAID FOR IN FULL.

4. If the invoice does not properly identify and describe the merchandise you intended to purchase, or if the materials delivered do not conform to the description on the invoice, or if the quantities do not match those on the face of the invoice, Purchaser is required to notify Seller within 48 hours thereof in writing.

5. In the event of any failure or omission on Purchaser's part to notify Seller of any variation in kind or quantity between the materials delivered and the description thereof on the invoice, as set forth in #4 above, it will be conclusive between Purchaser and Seller that the invoice properly identifies the materials purchased by Purchaser, and sold and delivered to Purchaser or for Purchaser's account, and the quantity thereof delivered to Purchaser and that the materials actually delivered by Seller to Purchaser conform to the description thereof.

6. Vendor retains title to all merchandise until fully paid for, and shall have the right to repossess such merchandise in the event of default or non-payment in accordance with these terms of sale. Furthermore, vendor reserves the right to refuse to make deliveries of any merchandise except for cash.

7. We assume no responsibility for delays due to fire, strikes, or other causes that are unavoidable or beyond our control, including those attributable to our source of supply.

8. Merchandise made to customer's specifications, or of special manufacture, or ordered especially for the customer, is sold subject to the condition that such order is non-cancelable.

9. A charge of 20 percent will be made for all materials returned. All returns must be in merchantable condition, in the same packaging and or wrap-per condition as when delivery thereof was made. Return of goods is at the manufacturers discretion.

10. Customer accepts primary liability for the payment of all merchandise shipped and/or billed hereon, and waives the right to use as a defense against payment - the obligation by the third party to pay for said merchandise. For the purpose of this clause, the "Customer" shall be the party or company who placed the order with the vendor, or at whose instigation and/or request the merchandise was shipped, however, the vendor's rights shall not be limited to such party or customer.

11. All claims relating to customer's liability, not covered by paragraphs #1, #5 and #6 above must be made in writing, by registered mail within ten (10) days of receipt of this invoice but in no event more than thirty (30) days following the date or it will be presumed that the materials as delivered were as set forth on the face hereof.

12. Customers who are charged New York State Tax shall be liable for the payment of New York State Sales Tax, and all other applicable taxes, whether or not charged by vendor, furthermore, customer agrees to reimburse vendor for all assessments, penalties, interest, and similar charges levied by the State of New York or any of its political subdivisions, department, or agents, or by any other taxing authority, arising out of related to, or in any way connected with customer's failure to fulfill its obligations under the terms of this clause and the New York State Sales Tax regulations, or any other applicable tax regulation(s) or law(s).

13. Purchaser agrees to have agent and facilities available to accept all deliveries of merchandise pursuant to orders placed with vendor, for shipment to customer's office, warehouse, jobsite, or any other designated location. Customer shall take full responsibility for freight, demurrage, hold-over, and similar charges arising out of, connected with, or relating to customer's failure to fulfill its obligations under the terms of this clause - regardless of the origin of the shipment of the consignor.

14. Purchaser shall hold vendor harmless in all actions, proceeding, and litigation arising out of charges brought against manufacturer(s) whose products are sold by vendor.

15. The Purchaser agrees to waive trial by jury in any action proceeding, or counter claim brought by either party hereto, of any matter arising out of, or in any way connected with, or pertaining to the sale of the installation, use or other disposition of the merchandise sold, or any claim for injury or damages arising therefrom.

16. IN THE EVENT THE TERMS OF THIS AGREEMENT ALTER OR VARY THE TERMS OF PURCHASER'S PURCHASE ORDER, THE TERMS OF THIS AGREEMENT SHALL PREVAIL UNLESS PURCHASER NOTIFIES US IN WRITING WITHIN 10 DAYS OF RECEIPT OF SELLER'S CONFIRMATION.

17. The rights of the vendor are not limited to the above.

18. All invoices are due upon receipt.

19. If any invoice is not paid within thirty (30) days of the date hereof, it shall accrue service charges at the rate of 2% per month until the day of payment.

20. In the event that Seller resorts to the legal action for the collection of any invoice, or any part hereof, Purchaser shall be liable for the payment of legal fees equal to ONE THIRD OF THE AMOUNT OF SAID INVOICE TOGETHER WITH ANY DISBURSEMENTS OR OTHER COSTS EXPENDED IN SUCH COLLECTION.

DATE

CUSTOMER

RVE-85-52D-20I-J-A2

Unit Performance

Design Conditions						
Elevation (ft)	Summer		Winter DB (F)	Supply (CFM)	Outdoor Air (CFM)	Exhaust Air (CFM)
	DB (F)	WB (F)				
23	88.9	75.0	3.5	5,375	5,375	4,800

Unit Specifications						
Qty	Weight (lb)	Cooling Type	Heating Type	Unit Installation	Unit ETL Listing	Furnace ETL Listing
2	4,685 (+/- 5%)	Packaged DX	Indirect Gas	Outdoor	UL\cUL 1995/60335-2-40	ANSI Z83.8 / CSA 2.6

Configuration			
Outdoor Air		Exhaust Air	
Intake	Discharge	Intake	Discharge
End	Bottom	Bottom	Side

ASHRAE 90.1 Compliance			
	ASHRAE 90.1 Min. Efficiency	Calculated Efficiency	Compliance
ISMRE2 (ASHRAE 90.1-2022)	5.0	7.2	✓
Enthalpy Recovery Ratio (%)	50	62.6	✓

Energy Recovery Performance									
Design Condition	Temperature (F)								Capacity Reduction (BTU/h)
	Outdoor Air		Supply Air		Return Air		Exhaust Air		
	DB	WB	DB	WB	DB	WB/RH	DB	WB	
Summer	88.9	75.0	79.9	67.6	75.0	62.5/50	84.9	71.5	157,219.0
Winter	3.5	1.7	45.4	38.9	72.0	55.8/35	22.6	21.8	243,229.0

Cooling Specifications							
Type	Total Capacity (MBH)	Sensible Capacity (MBH)	Lead Compressor Type	Coil (DB/WB)		Reheat	
				EAT (F)	LAT (F)	Capacity (MBH)	LAT (F)
Packaged DX	268.5	165.0	Inverter Scroll	79.9 / 67.6	50.7 / 50.6	185.3	82.7

Heating Specifications								
Type	Gas Type	Input (MBH)	Output (MBH)	Temperature Rise		Turndown	Performance	
				Min (F)	Max (F)		EAT (F)	LAT (F)
Indirect Gas	Natural	300.0	243.0	3.0	42.0	12:1	45.4	87.2

Motor Specifications						
Motor	Qty	Operating Power (hp)	Size (hp)	Enclosure	Efficiency	RPM
Supply	1	5.28	7-1/2	ODP	PE	1770
Exhaust	1	3.04	5	ODP	PE	1750

Electrical Specifications					
Power Supply	Rating (V/C/P)	MCA (A)	MOP (A)	FLA (A)	Fan Power (W/CFM)*
Unit	208/60/3	129.9	150.0	120.3	1.154

*Fan Power (W/CFM) = (Supply BHP + Exhaust BHP) / Supply CFM

Construction Features And Accessories

Unit	
Unit Installation - Outdoor	Std
Unit Construction - Double Wall	Std
Insulation - 2 inch 2.4# R13 foam	Std
Corrosion Resistant Fasteners	Std
Hinged Access	Std
Factory Wired Non-Fused Disconnect Switch	Std
Direct Drive Plenum Blower & Motor Assemblies	Std
Factory Wired VFDs	Std
Unit Finish - Permatector, Concrete Gray (RAL 7023)	X
Stainless Steel Condensate Drain Pan and Connection	Std
Condensate Drain Trap	Std
Short Circuit Current - 5 kA	Std
Energy Recovery Device - Polymer Wheel w/ Silica Gel Desiccant	Std
Controls	
Unit Controls - Full Control	Std
Internally Mounted Control Center with 24 VAC control transformer(s) and control circuiting fusing	Std
BMS Protocol - BACNetMSTP	X
BMS Monitoring Points	
Supply Fan Control - CO2 - By Factory	X
Exhaust Fan Control - Space Static Pressure by Factory	X
Economizer Control - Temp./Enthalpy	X
Exhaust Fan Only Power	
Web-Based User Interface	Std
Energy Wheel Economizer Control - Modulating Wheel, OA Temp Setpoint w/VFD Wheel	X
Energy Wheel Rotation Sensor	Std
Damper Control - 100% OA-No Recirculation	X
Unoccupied Recirc Mode	
Control Accessories	
Remote Display - w/150 ft cord	X
Dirty Filter Sensor(s)	
Airflow Monitor	
Room Thermostat - Space Temp and RH	X
Phase/Brownout Protection	Std
Economizer Fault Detection Diagnostics	

Accessories	
Frost Control Modulating Wheel - Modulating Wheel	X
Outdoor Air Damper - Low Leakage	X
Return Air Damper - Low Leakage	X
Roof Curb - GKD - 61.6/156.6-G14	X
Supply Air Filters - 2" Merv 13, 3-16x20x2, 6-20x20x2	X
Service Outlet - Factory mounted and wired	X
Piping Vestibule	
Service Lights	
Condensate Overflow Switch	X
Spare Filters	
Exhaust Discharge Gravity Backdraft Damper	X
ElectroFin Coil Coating	
Motor Shaft Grounding	
Bipolar Ionization	
Smoke Detector(s), Both - Shipped Loose	X
Barometric Relief Damper	
UV Lights	
Return Air Filters - 2" Merv 8, 4-16x25x2	Std
Outdoor Air Filters - 2" Merv 8, 4-16x25x2	Std
Furnace Control - 12:1 Modulating	X
Spare Energy Wheel Belt	
Spare Energy Wheel Segments	
Energy Wheel Bypass Damper	
Power Venting	Std
Hail Guards	
Warranty Options	
Unit Warranty - 18 Months (Std.)	Std
Energy Wheel Warranty - 5 Yrs Less Motor	Std
Compressor Warranty - 18 Months (Std.)	Std
Furnace HX Warranty - 25 Yrs.	Std

Standard Option	Std
Not Included	
Included	X

Notes	
Outdoor Air Damper supplied is low leakage, motorized VCD-23 (leakage rate of 3 CFM/ft ² @ 1 in. wg), Class 1A	
Return Air Damper supplied is low leakage, motorized VCD-23 (leakage rate of 3 CFM/ft ² @ 1 in. wg), Class 1A	
Smoke Detector(s) - sampling tube included sized for duct widths of 1-2 ft.	

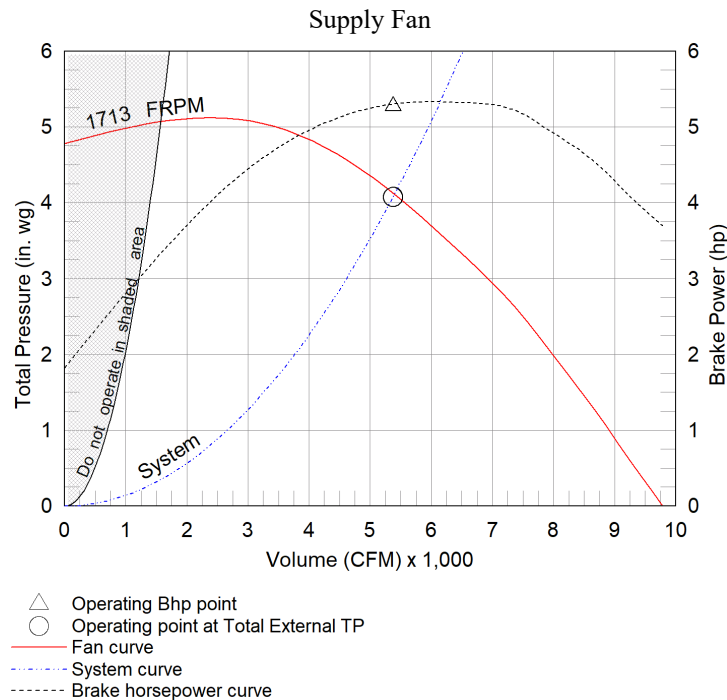
Supply Fan Charts And Performance

Supply Fan Performance									
Total Volume (CFM)	External SP (in. wg)	Total SP (in. wg)	RPM	Operating Power (hp)	Motor		Fan		
					Qty	Size (hp)	Qty	Type	Drive-Type
5,375	1.5	4.077	1713	5.28	1	7-1/2	1	Plenum	Direct

Pressure Drop (in. wg)							
Weatherhood	Filter	Damper	Cooling	Heating	External	Energy Wheel	Total
0.16	0.142	0.05	0.285	0.541	1.5	1.37	4.077

Sound Performance in Accordance with AMCA									
Sound Power by Octave Band								Lwa	dBA
62.5	125	250	500	1000	2000	4000	8000		
88	89	95	82	79	73	72	70	88	77
								Sones	
								28	

*Energy Wheel pressure drop shown in above table also accounts for pressure drop across MERV8 OA filter



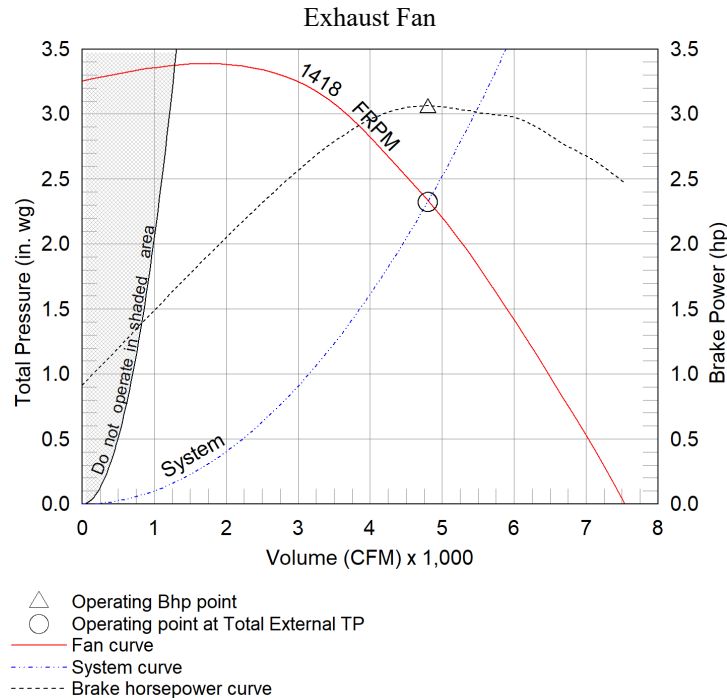
Exhaust Fan Charts And Performance

Exhaust Fan Performance										
Mode	Total Volume (CFM)	External SP (in. wg)	Total SP (in. wg)	RPM	Operating Power (hp)	Motor		Fan		
						Qty	Size (hp)	Qty	Type	Drive-Type
Normal	4,800	1.25	2.325	1418	3.04	1	5	1	Plenum	Direct

Pressure Drop (in. wg)								
Mode	Weatherhood	Filter	Damper	Cooling	Heating	External	Energy Wheel	Total
Normal	0.06	-	-	-	-	1.25	1.02	2.325

Sound Performance in Accordance with AMCA											
Mode	Sound Power by Octave Band								Lwa	dBA	Sones
	62.5	125	250	500	1000	2000	4000	8000			
Normal	91	82	82	72	66	64	63	64	77	66	16

*Energy Wheel pressure drop shown in above table also accounts for pressure drop across MERV8 return air filter

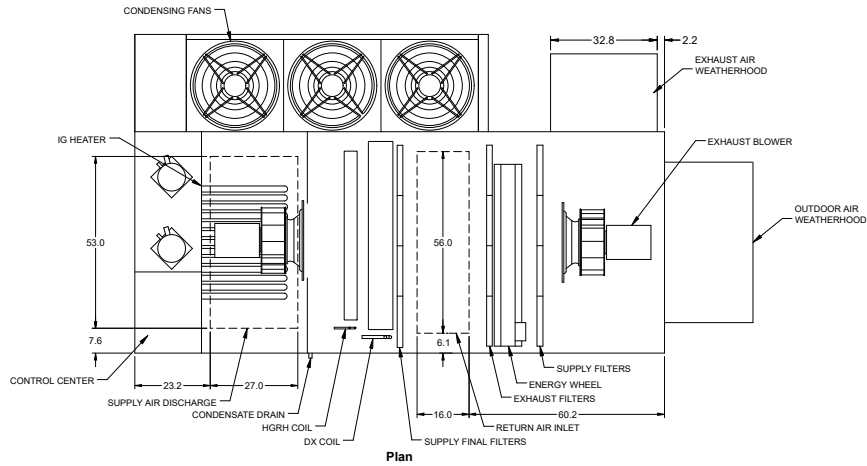


Radiated Sound

Position A

Dimensional Overview

Position D



Position B

Position C

"E" is the
Top Plane

Supply Air Flow Nominal, Largest Tonnage Condensing Section Available, PDX units only

Radiated Sound Levels										
Plane	Octave Bands (Lw)								Plane Lw	Plane LwA
	1	2	3	4	5	6	7	8		
A	80	87	82	80	77	72	68	62	90	82
B	74	80	73	67	64	59	55	48	82	71
C	82	77	73	69	67	61	54	46	84	72
D	71	80	74	72	69	63	58	52	82	74
E	77	86	82	78	76	71	66	61	89	81
Total	86	91	86	83	80	75	71	65	94	85

AMCA 320-07 - Laboratory Methods of Sound Testing of Fans Using Sound Intensity

Tests conducted in accordance with this standard.

Free field measurement plane created 1 foot from unit on all sides and top.

Sound Intensity measured in Watts/m².

Sound data converted to Sound Power (Lw) for the chart above.

A-Weighted Sound Power was determined using AMCA Standard 301-90 Clause 9.1.

Plane E sound data was measured above the top plane of the unit.

Cooling Performance

Cooling Specifications									
Nominal Tonnage	Entering Air (F)		Leaving Air (F)		Capacity (MBH)		Reheat		Condensing Ambient Temp (F)
	DB	WB	DB	WB	Total	Sensible	Capacity (MBH)	LAT (F)	
20.0	79.9	67.6	50.7	50.6	268.5	165.0	185.3	82.7	88.9

Coil Information									
PDX Coil Model	Fins Per Inch	Rows Deep	Face Vel. (ft/min)	Coil PD (in. wg)	Refrigerant	Refrig. Velocity (ft/min)	Face Area (ft2)	Suction Temp (F)	
DR38S06H14-58x56-LH	14	6	238	0.285	R-454B	1,267	22.6	47.4	

Compressor Details					
Lead Compressor Type	Compressor Qty	Compressor RLA/MRC (A)		Compressor LRA (A)	
		Comp. #1	Comp. #2	Comp. #1	Comp. #2
Inverter Scroll	2	38.4	33.3	NA	255

A2L Installation Requirement - UL 60335-2-40		
Largest Circuit Charge	Minimum Circulation Airflow	Minimum Total Conditioned Room Area
48.6lb / 22.06kg	1,318 CFM	730 ft2

Local codes and standards may have requirements regarding the installation of A2L refrigerants in addition to manufacturing instructions provided for listed and labeled equipment.

Unit Details
Refrigerant charges provided by the factory are approximate and may require adjustment in the field
Hermetic scroll type compressors
Compressors mounted on neoprene vibration isolation
Crankcase heater on staged compressor
Electronic expansion valve on lead circuit, thermostatic expansion valve on staged circuit
Stainless steel double sloped drain pan
Moisture-indicating sight glass
Service/charging valves
Refrigerant high pressure switch (manual reset)
Liquid-Line filter drier
Leak detection sensors
Multiple low sound condensing fans with Lead ECM condensing fan for modulating head pressure control
Inverter scroll compressor
Refrigerant low pressure switch (auto reset)
Unit cannot be mounted in an enclosed space.

Important Notes:
Capacity is based on incoming voltage selected. If incoming power varies it may affect the capacity of your selection.

Heating Performance

Heating Specifications								
Type	Gas Type	Input (MBH)	Output (MBH)	Temperature Rise		Turndown	Performance	
				Min (F)	Max (F)		EAT (F)	LAT (F)
Indirect Gas	Natural	300.0	243.0	3.0	42.0	12:1	45.4	87.2

Unit Details
ANSI standard Z83.8 and CSA 2.6
High Thermal efficiency
Direct spark ignition
3/4" Gas Connection
At least 6 in. wg of natural gas pressure (14 in. wg for LP) is required at the units gas connection in order to achieve maximum performance
Power Venting
24 Volt Control Power
Stainless Steel heat exchange tubes
Unit controller maximum allowable supply discharge air set point is 100F (37.8C)
Discharge temperature assumes proper energy wheel operation and maintenance.

Energy Recovery Summer Performance

Outdoor Air		Supply Air	
Dry Bulb (F)	88.9	Dry Bulb (F)	79.9
Wet Bulb (F)	75.0	Wet Bulb (F)	67.6
Specific Humidity (gr/lb)	109	Specific Humidity (gr/lb)	82
Enthalpy (BTU/lb)	38.5	Enthalpy (BTU/lb)	32.0
Exhaust Air		Return Air	
Dry Bulb (F)	84.9	Dry Bulb (F)	75.0
Wet Bulb (F)	71.5	Rel. Humidity (%)	50
Specific Humidity (gr/lb)	94	Specific Humidity (gr/lb)	65
Enthalpy (BTU/lb)	35.2	Enthalpy (BTU/lb)	28.1

Design Air Flow Conditions

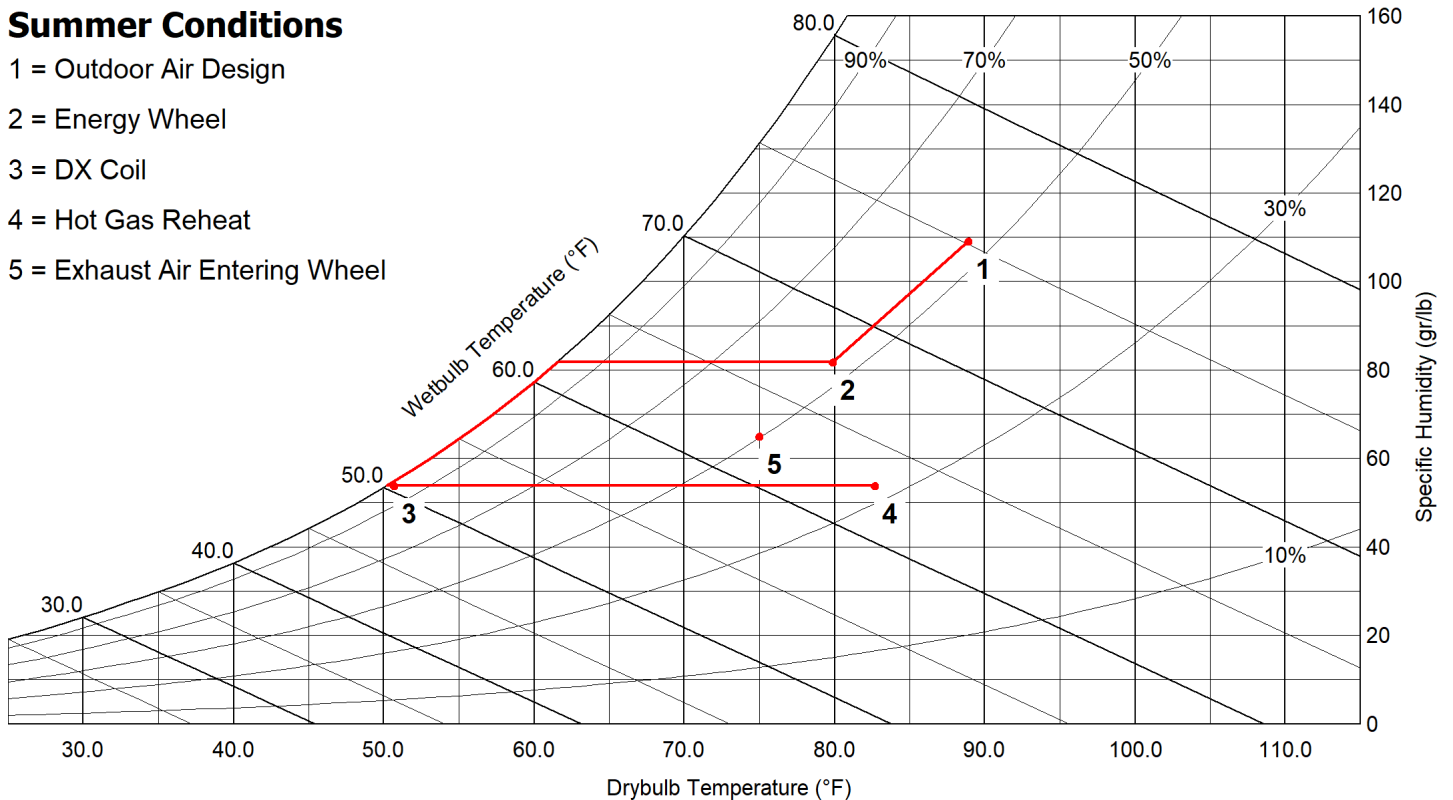
OA Volume (CFM)	ASHRAE 90.1 OA Enthalpy Recovery Ratio	EA Volume (CFM)	EA Wheel Effectiveness
5,375	62.6	4,800	68

Outdoor Air Cooling Reduction

OA Load w/o Energy Recovery		OA Load with Energy Recovery		Equipment Reduction (tons)
(BTU/h)	(tons)	(BTU/h)	(tons)	
251,550.0	20.96	94,331.0	7.86	13.10

Summer Conditions

- 1 = Outdoor Air Design
- 2 = Energy Wheel
- 3 = DX Coil
- 4 = Hot Gas Reheat
- 5 = Exhaust Air Entering Wheel



Energy Recovery Winter Performance w/out Preheater

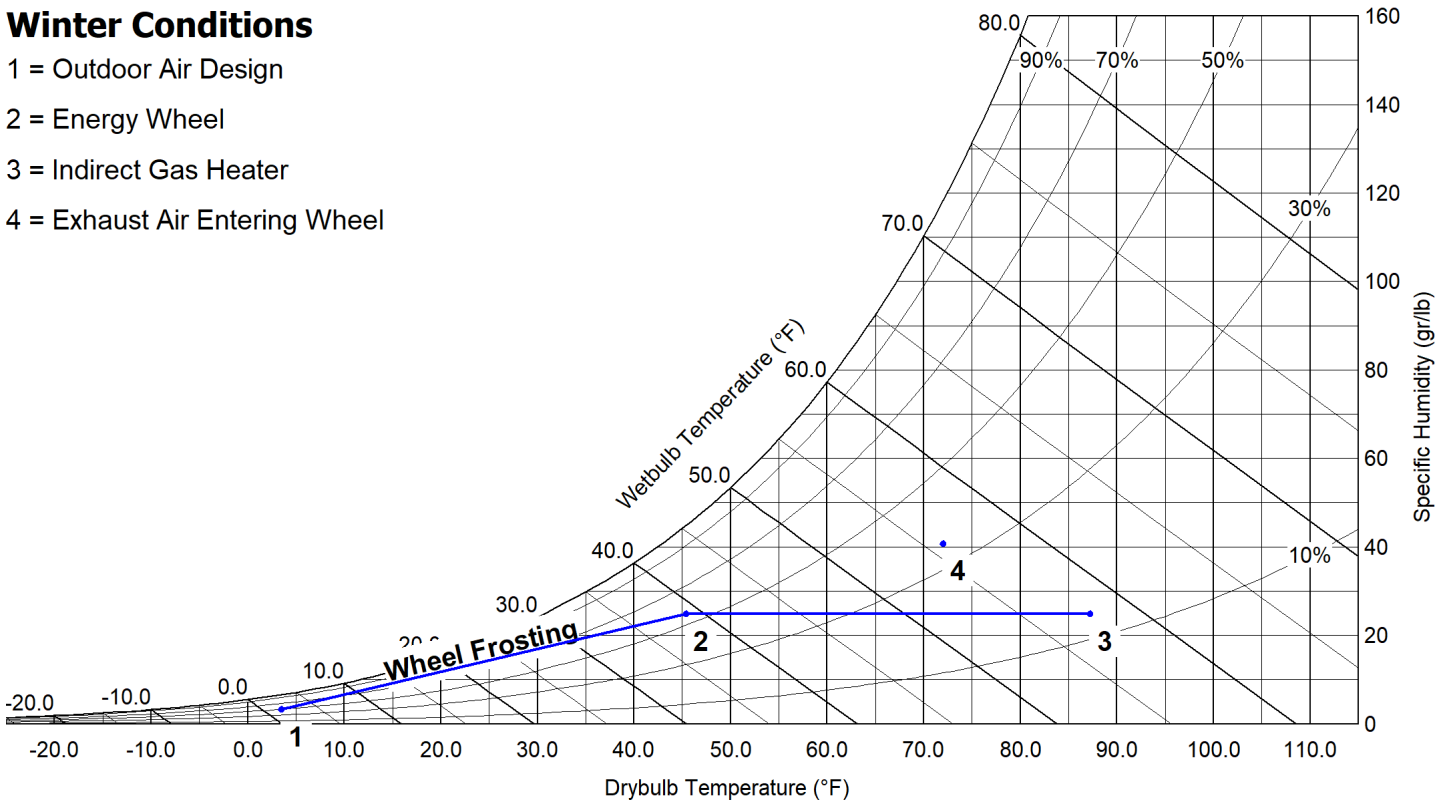
Outdoor Air		Supply Air	
Dry Bulb (F)	3.5	Dry Bulb (F)	45.4
Wet Bulb (F)	1.7	Wet Bulb (F)	38.9
Specific Humidity (gr/lb)	3	Specific Humidity (gr/lb)	25
Enthalpy (BTU/lb)	1.3	Enthalpy (BTU/lb)	14.7
Exhaust Air		Return Air	
Dry Bulb (F)	22.6	Dry Bulb (F)	72.0
Wet Bulb (F)	21.8	Rel. Humidity (%)	35
Specific Humidity (gr/lb)	15	Specific Humidity (gr/lb)	41
Enthalpy (BTU/lb)	7.8	Enthalpy (BTU/lb)	23.7

Design Air Flow Conditions			
OA Volume (CFM)	ASHRAE 90.1 OA Enthalpy Recovery Ratio	EA Volume (CFM)	EA Wheel Effectiveness
5,375	60	4,800	70.3

Outdoor Air Heating Reduction			
OA Load w/o Energy Recovery (BTU/h)	OA Load with Energy Recovery (BTU/h)	Equipment Reduction (BTU/h)	Sensible Effectiveness (%)
397,642.0	154,413.0	243,229.0	71.3

Winter Conditions

- 1 = Outdoor Air Design
- 2 = Energy Wheel
- 3 = Indirect Gas Heater
- 4 = Exhaust Air Entering Wheel



AHRI Performance Ratings

Energy Recovery Performance Rating in accordance with AHRI Standard 1060 (I-P)

Rated Airflow (SCFM)		Net Supply Airflow (SCFM)	EATR (%)	OACF	Pressure Drop (in. wg)		Purge Angle (degrees)
Leaving Supply	Entering Exhaust				Supply	Exhaust	
4,824	4,951	5,526	2.7	1.02	0.79	0.70	0

Thermal Effectiveness Ratings

Enthalpy Recovery		Sensible Effectiveness		Latent Effectiveness		Total Effectiveness	
Summer	Winter	Summer	Winter	Summer	Winter	Summer	Winter
62.6	60	70.8	71.3	66.7	67.2	68	70.3

Note(s)

Summer Design Conditions:

Certified in accordance with the AHRI ERV Certification Program, which is based on AHRI Standard 1060. Certified units may be found in the AHRI Directory at www.ahridirectory.org.

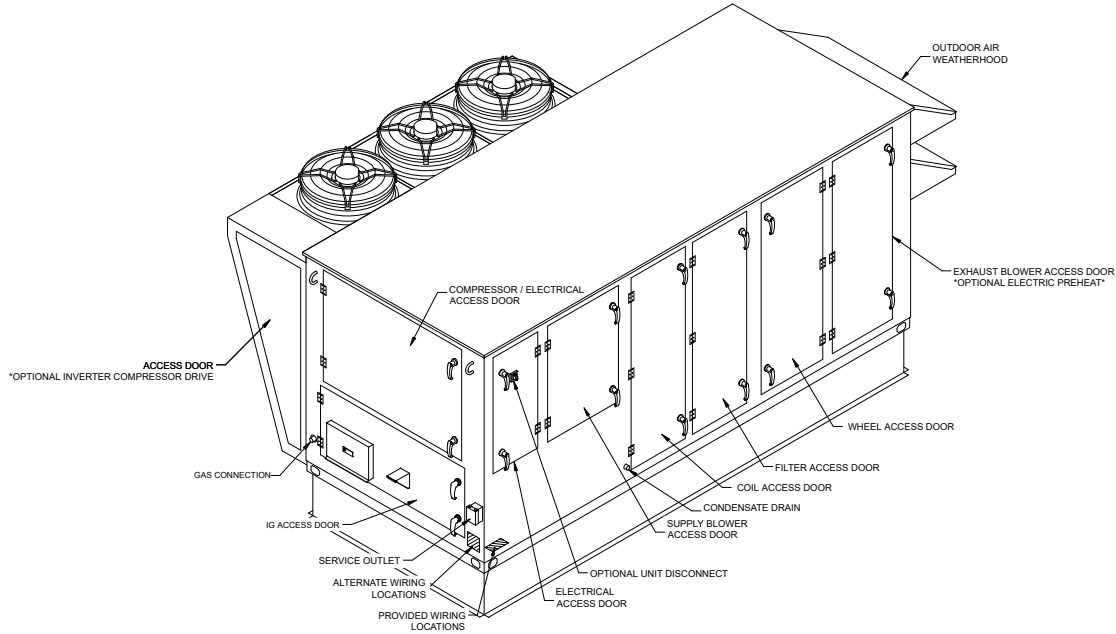


Winter Design Conditions:

Application Rating is outside the scope of the AHRI ERV certification Program but is rated in accordance with AHRI Standard 1060.

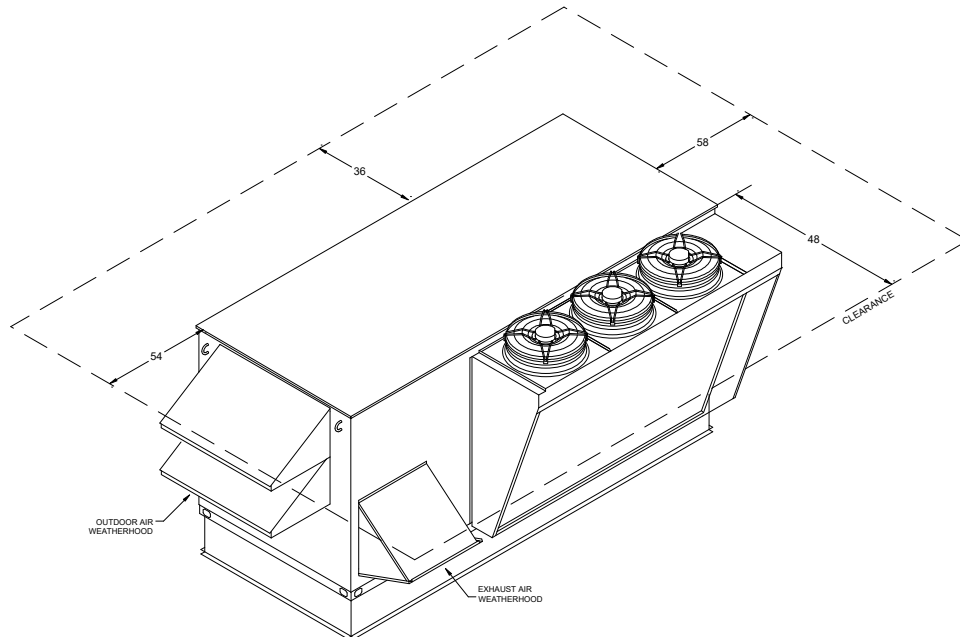
Isometric Drawings

Component Layout



Back Right Isometric

Service Clearances

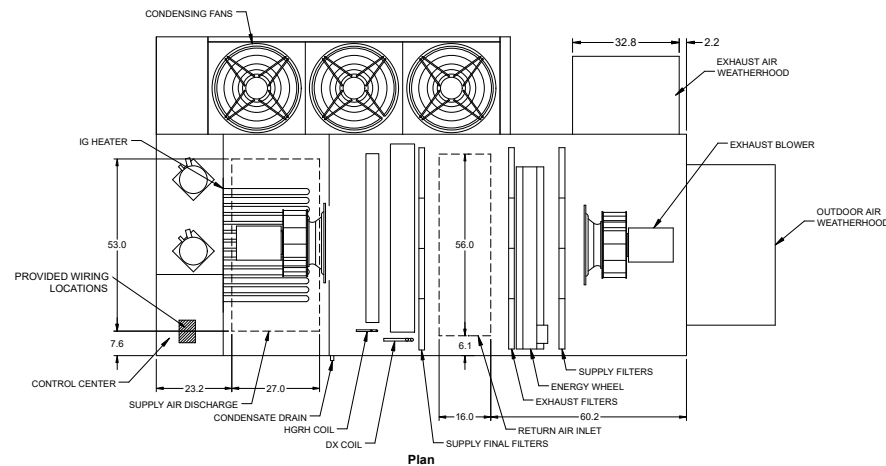
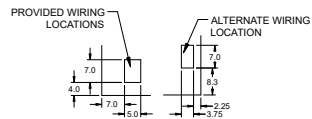


Front Left Isometric

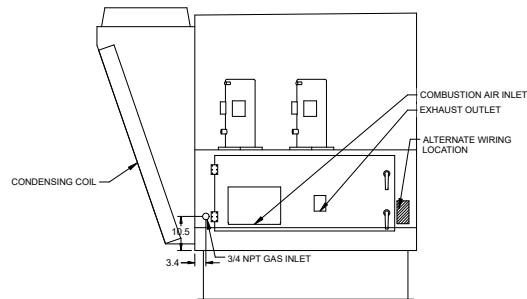
Overview Drawings

Dimensional Overview

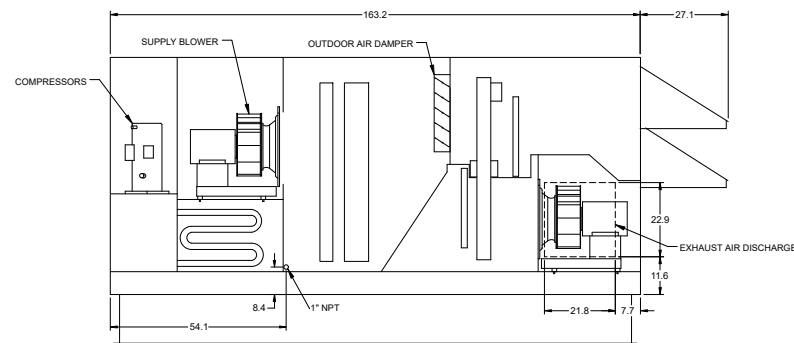
Electrical Connections



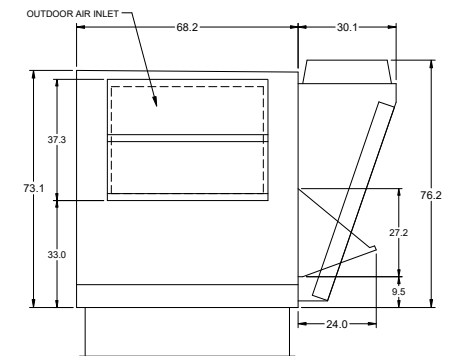
Plan



Left End

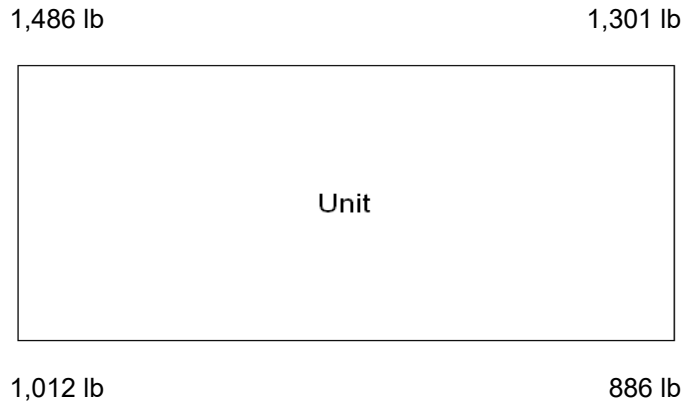


Elevation



Right End

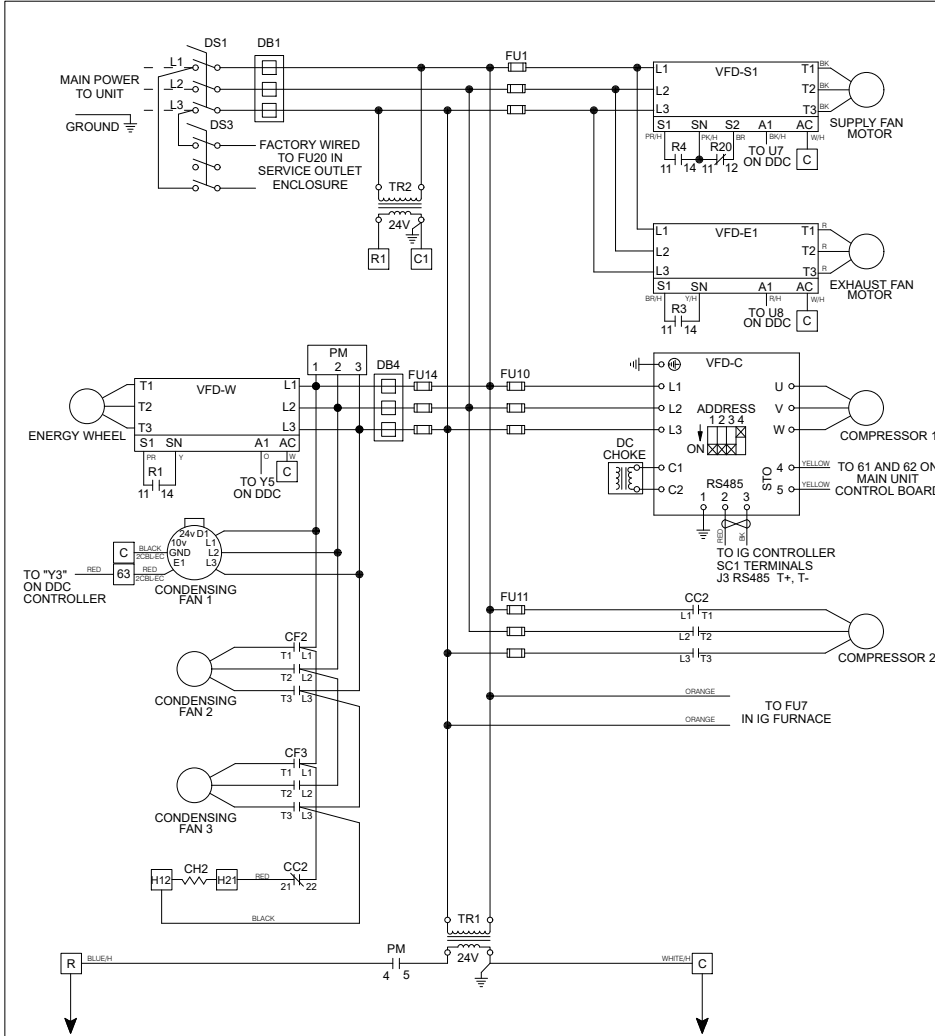
Unit Corner Weights



Note

Estimated corner weights are shown looking down on unit and the outside air intake will be on the right. Weights are applied at the base of the unit. Images not drawn to scale.

Wiring Diagram



Wiring Diagram Code:

G23N2E1BF02G25X00HJ23H0630BH31

CAUTION

UNIT SHALL BE GROUNDED IN ACCORDANCE WITH N.E.C.
POWER MUST BE OFF WHILE SERVICING.

NOTES

USE COPPER CONDUCTORS ONLY
60° C FOR TERMINALS RATED LESS THAN 100 AMPS.
75° C FOR TERMINALS RATED 100 AMPS OR MORE.
FIELD CONTROL WIRING RESISTANCE SHOULD
NOT EXCEED 0.75 OHM.

FIELD WIRED - - - - -
FACTORY SUPPLIED AND WIRED _____

WIRE COLOR CODE

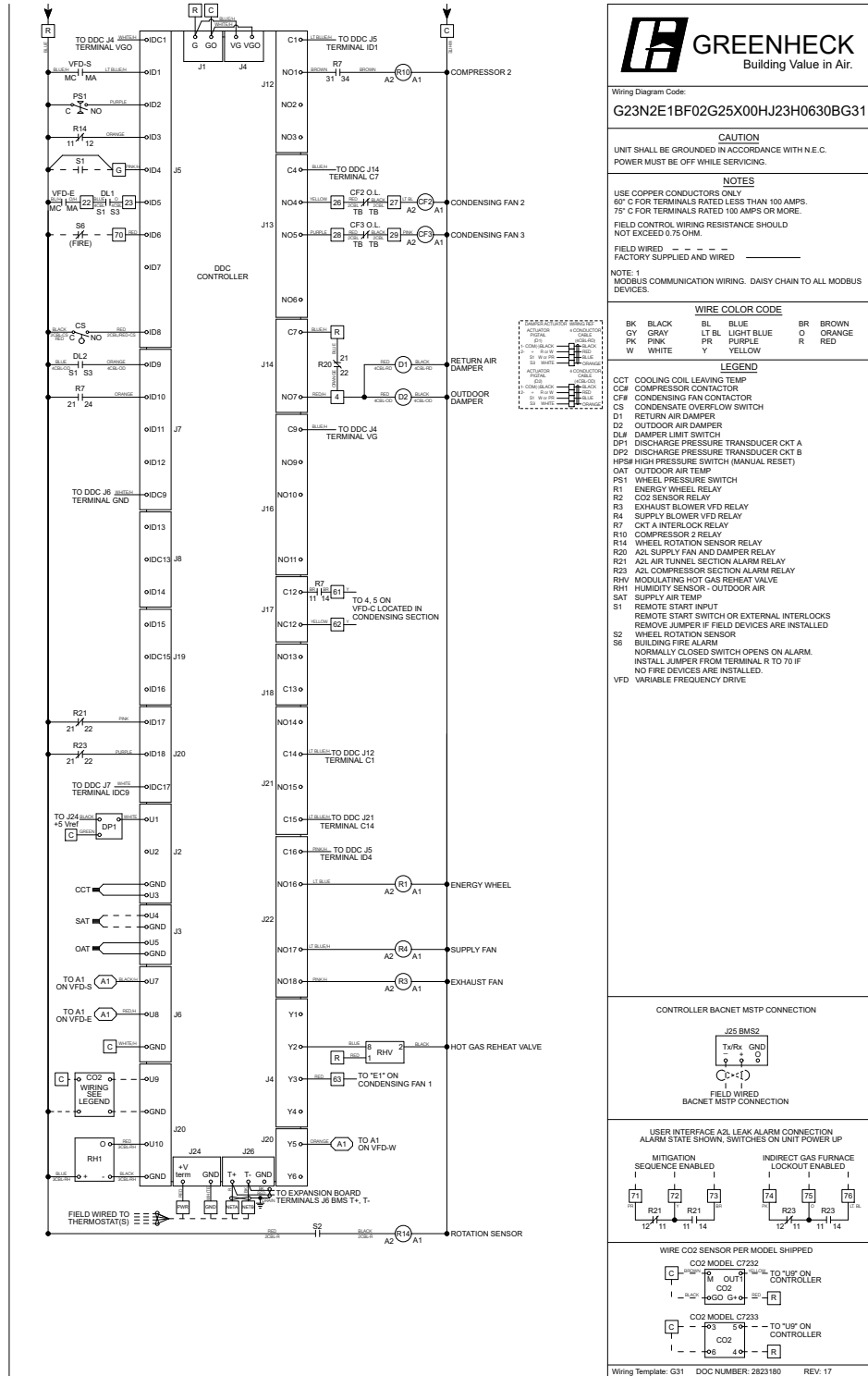
BK	BLACK	BL	BLUE	BR	BROWN
GY	GRAY	LT BL	LIGHT BLUE	O	ORANGE
PK	PINK	PR	PURPLE	R	RED
W	WHITE	Y	YELLOW		

LEGEND

CC#	COMPRESSOR CONTACTOR
CF#	CONDENSING FAN CONTACTOR
CH#	COMPRESSOR SUMP HEATER
DB#	POWER DISTRIBUTION BLOCK
DS	DISCONNECT SWITCH
FU#	FUSES
PM	PHASE VOLTAGE MONITOR
R1	ENERGY WHEEL RELAY
R3	EXHAUST BLOWER VFD RELAY
R4	SUPPLY BLOWER VFD RELAY
R20	A2L SUPPLY FAN AND DAMPER RELAY
TR#	TRANSFORMER
VFD	VARIABLE FREQUENCY DRIVE

Wiring Template: H31 DOC NUMBER: 2823180 REV: 17

Wiring Diagram 2



Wiring Diagram 3



GREENHECK
Building Value in Air.

Wiring Diagram Code:

G23N2E1BF02G25X00HJ23H0630BL31

CAUTION

UNIT SHALL BE GROUNDED IN ACCORDANCE WITH N.E.C.
POWER MUST BE OFF WHILE SERVICING.

NOTES

USE COPPER CONDUCTORS ONLY
60° C FOR TERMINALS RATED LESS THAN 100 AMPS.
75° C FOR TERMINALS RATED 100 AMPS OR MORE.
FIELD CONTROL WIRING RESISTANCE SHOULD NOT EXCEED 0.75 OHM.
FIELD WIRED - - - - -
FACTORY SUPPLIED AND WIRED _____

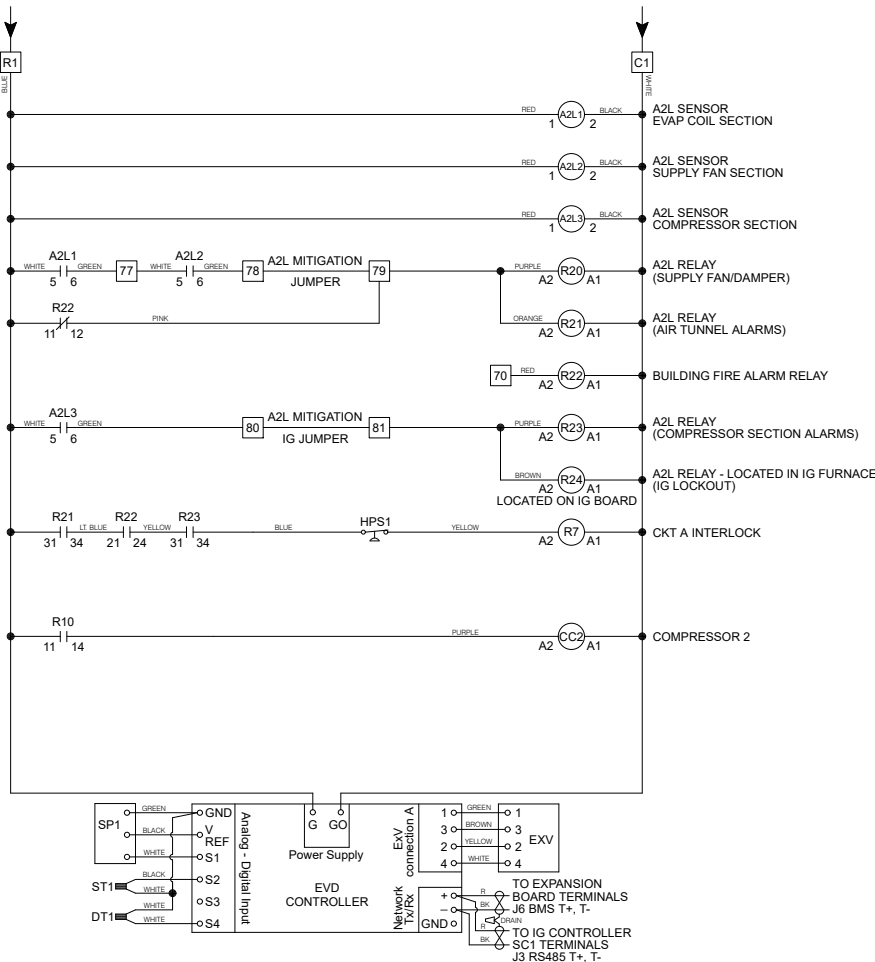
WIRE COLOR CODE

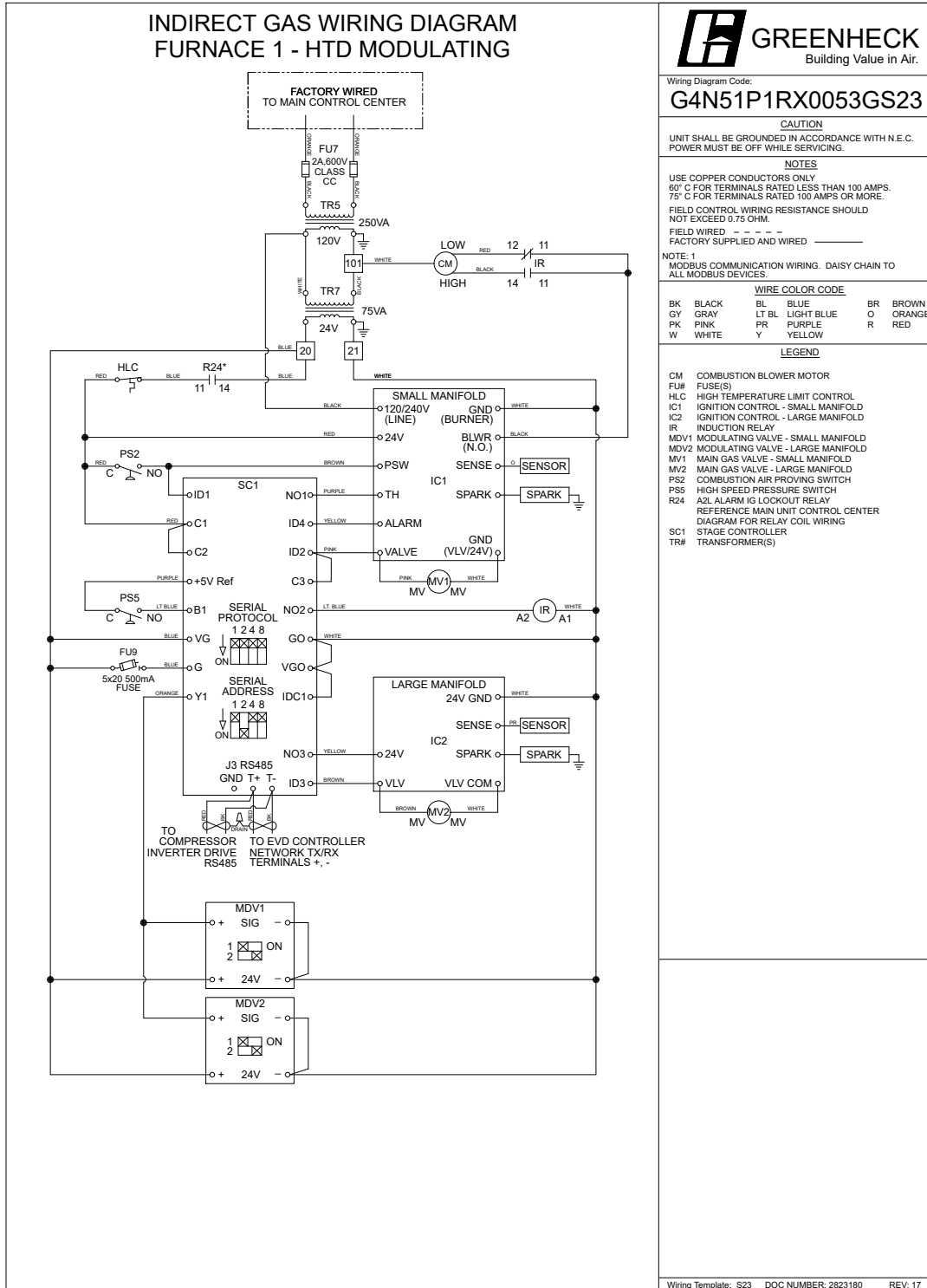
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GY	GRAY	LT BL	LIGHT BLUE	O	ORANGE
PK	PINK	PR	PURPLE	R	RED
W	WHITE	Y	YELLOW		

LEGEND

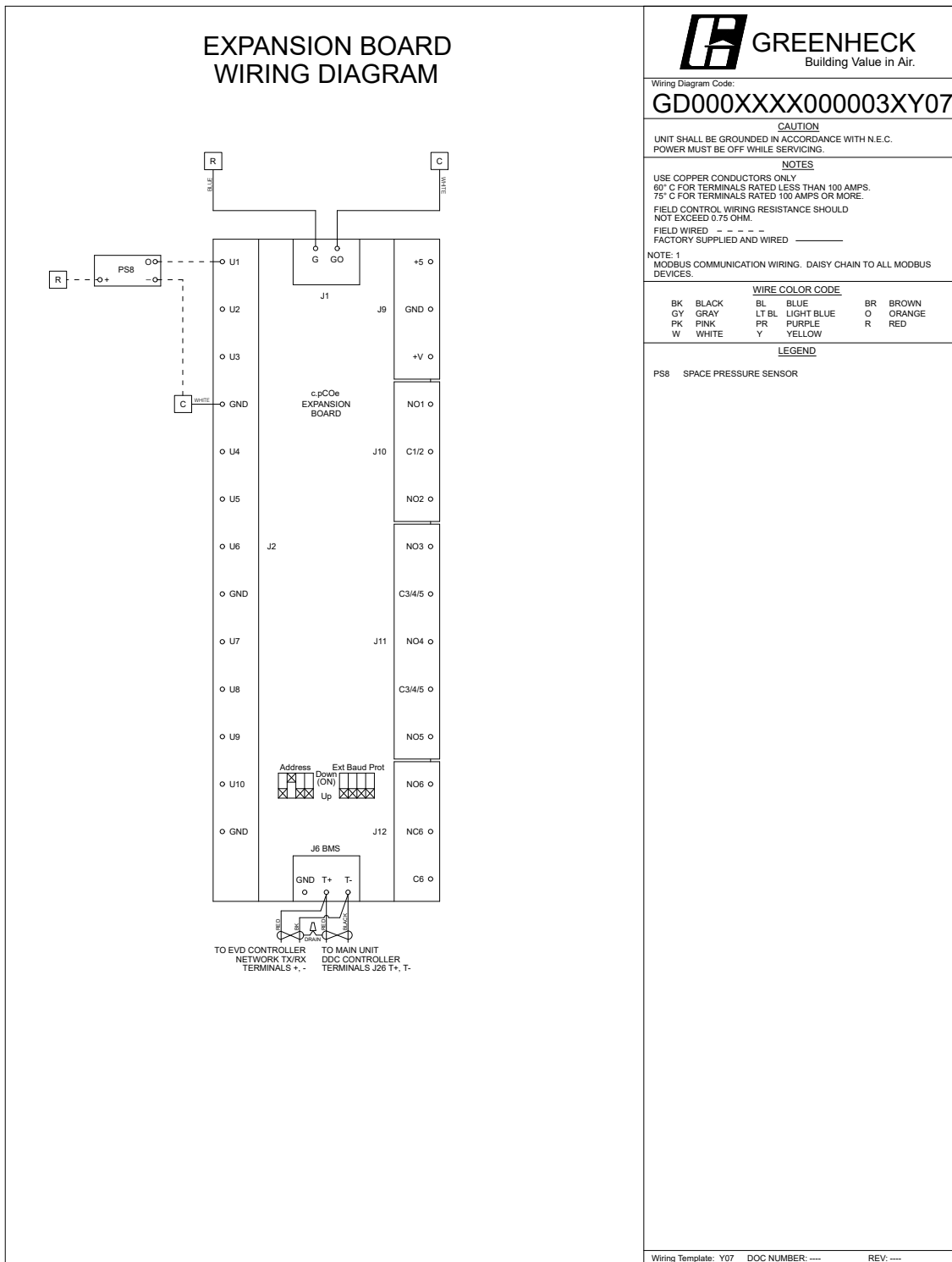
A2L# A2L REFRIGERANT LEAK SENSOR
CC# COMPRESSOR CONTACTOR
DT1 DISCHARGE LINE TEMP CKT A
EXV ELECTRONIC EXPANSION VALVE
R7 CKT A INTERLOCK RELAY
R10 COMPRESSOR 2 RELAY
R20 A2L SUPPLY FAN AND DAMPER RELAY
R21 A2L AIR TUNNEL SECTION ALARM RELAY
R22 BUILDING FIRE ALARM RELAY
R23 A2L COMPRESSOR SECTION ALARM RELAY
R24 A2L ALARM IG LOCKOUT RELAY
RELAY LOCATED ON IG FURNACE CONTROL CENTER BOARD
SP1 SUCTION PRESSURE TRANSDUCER CKT A
ST1 SUCTION LINE TEMP CKT A
VFD VARIABLE FREQUENCY DRIVE

Wiring Template: L31 DOC NUMBER: 2823180 REV: 17





Monitoring Points Wiring Diagram



Factory Controller Sequence of Operation

FACTORY CONTROLLER: Controller shall be provided with required sensors and programming for rooftop unit. Controller shall be factory programmed, mounted and tested. Controller shall have a LCD readout for changing set points and monitoring unit operation.

UNIT START COMMAND (Unit will be enabled to start once a jumper is placed between R to G):

- Factory mounted and wired outdoor air damper actuator is powered
- Return/Exhaust air damper actuator is powered.
- Exhaust fan starts after a (adj.) delay.
- Supply fan starts after a (adj.) delay.
- Tempering options and energy wheel option to function as described below.

UNIT STOP COMMAND (OR DE-ENERGIZED):

- Supply fan, exhaust fan, energy wheel and tempering options de-energized.
- Outdoor air damper actuator is spring return close.
- Return air damper is spring return close.

OCCUPIED/UNOCCUPIED MODES: Shall be based on a 7-day time clock internal to the controller. The schedule shall be set by the end user. When a user initiates an override input, the controller will switch from unoccupied to occupied mode. The controller will return to the scheduled occupied/unoccupied mode after the override time has expired. If internal time clock is disabled, a remote contact or a BMS can control the occupied/unoccupied mode.

Occupied Mode:

- Damper control per below.
- Energy wheel control per below.
- Exhaust fan ON.
- Supply fan ON.
- Heating per below.
- Cooling per below.

Unoccupied Mode (Unit Off): Unit remains off when in unoccupied mode.

- Supply fan OFF
- Exhaust fan OFF
- Tempering OFF
- Outdoor air damper closed.
- Return damper closed.

MORNING WARMUP/COOL DOWN: Prior to occupancy, the unit will run using the warmup or cool down sequence until the occupied set point is achieved. The heating or cooling mode must not be locked out and the space temperature is below or above set point by the unoccupied hysteresis (adj.) (This Sequence must be field configured.)

SUPPLY BLOWER SEQUENCE: The supply blower is provided with a factory mounted variable frequency drive. The supply blower speed will be controlled with the following sequence. Minimum supply fan turndown is 50% of the design maximum operation.

CO2 Sensor: The supply blower speed is modulated based upon the signal from a CO2 Sensor (factory provided, field mounted and wired). The controller will modulate the supply fan based upon a comparison of the CO2 set point (adj.) to the actual CO2 levels reported from the sensor. **(Mechanical high static protection cutoffs must be installed by others to protect the system and equipment from over-pressurization. The manufacturer does not assume responsibility for this.)**

EXHAUST BLOWER SEQUENCE: The exhaust blower will operate at a constant speed set point (adj.) during operation.

Space Static Pressure by Factory: The exhaust blower is modulated based upon the signal from a building pressure sensor. The controller will modulate the exhaust fan based upon a comparison of the building static pressure set point (adj.) and the building static pressure level reported from the sensor.

COOLING SEQUENCE: The cooling is controlled to maintain the supply temperature set point. The mechanical cooling will be locked out when the outside air is < 55 F (adj.).

Packaged DX Cooling (Inverter Scroll): The controller will provide a modulating signal for cooling. From 0-50%, the inverter scroll will be controlled to maintain discharge temperature. From 50-100% the second stage will be on in combination with the inverter scroll compressor to maintain the discharge temperature. The electronic expansion valve will modulate to maintain 8F of superheat.

Modulating Hot Gas Reheat Sequence: During dehumidification the modulating HGRH is controlled to maintain the supply temperature set point.

Modulating Head Pressure Control: Lead condenser fan will have an EC motor and will modulate to maintain a head pressure set point.

REHEAT SEQUENCE: While the unit is in dehumidification mode the outdoor air will be reheated via Modulating Hot Gas Reheat for space neutral applications.

Modulating Hot Gas Reheat: The controller will modulate the hot gas reheat valve with a 0-10 V signal to maintain the supply temperature set point (adj.).

HEATING SEQUENCE: The heating is controlled to maintain the supply temperature set point. The heating will be locked out when the outside air is > 80 F (adj.).

Indirect Gas Furnace: The controller will modulate the indirect gas furnace to maintain the supply temperature set point (adj.).

TEMPERATURE CONTROL SEQUENCE: The unit will maintain the supply air discharge setpoint per the following. Adjustable locally or by BMS.

Room Reset: The controller will reset the supply air temperature set point to maintain the room temperature set point (adj.).

Space Setpoint Control: The supply setpoint will adjust between minimum (adj.) and maximum (adj.) limits, to satisfy the desired space temperature setpoint. Adjustable locally or by BMS.

BUILDING FREEZE PROTECTION: If the supply air temperature drops below 35 F (adj.) for 300s (adj.), the controller will de-energize the unit and activate the alarm output.

TEMPERATURE PROTECTION: The controller will enable the supply fan to modulate down to help the unit keep up with heating demand in the event of wheel failure or the unit operating outside design conditions. (This can be enabled under the manufacturer menu in the controller)

UNIT LEAK DETECTION AND MITIGATION: The unit will be equipped with refrigerant leak detection sensors. These sensors along with the following sequence of operation are required per UL60335-2-40.

Refrigerant Leak Detected In Air Tunnel: If a refrigerant leak is detected in the air tunnel, the supply fan will operate at minimum airflow requirement, outside air damper will be opened, powered exhaust fan will operate based configured sequence, energy recovery wheel will operate if commanded by normal unit sequencing, and compressors are disabled to reduce leakage rate. This operation is required in order to move stagnant refrigerant from within the unit, duct, and space ensuring proper dilution of the refrigerant. This operation is required even if the unit is called to be off. After leak detection is cleared, the unit will go back to normal operation. Exception to this operation is when the unit is receiving an active fire alarm signal at the unit controller. If unit controller is receiving a fire alarm input, the unit will not operate the leak mitigation supply fan sequence.

Dry alarm contacts available to allow the building (by others) to perform external mitigation actions when necessary. These by other external actions include opening of zone dampers in the ductwork, disabling duct mounted electric resistance heaters, and/or enabling additional mechanical ventilation if required per ASHRAE 15.

Refrigerant Leak Detected In Compressor Compartment: If a refrigerant leak is detected in the compressor compartment and the unit is configured with an indirect gas furnace, the furnace will be disabled while leak detection is active. After leak detection is cleared, the unit will go back to normal operation.

ENERGY WHEEL FROST CONTROL: Frost control for the energy wheel is enabled when frost is present on the wheel; based on the outside air temperature and the pressure drop across the wheel. If the outdoor air temperature is below 5 F adj. and the differential pressure across the wheel is about 1.5", adj. frost control will enable.

Wheel VFD (Modulate Wheel): When frosting is occurring, the VFD modulates the wheel down to a slow rotational speed to defrost wheel. Once either the pressure drop decreases below the pressure switch set point, or the outdoor air temperature increases about the temperature set point, the unit will resume normal operation.

ECONOMIZER SEQUENCE: When the application requires cooling, and the outdoor air conditions are suitable for free cooling, the controller will modulate the energy wheel speed to maintain the discharge temperature set point. If the energy wheel speed modulates to the economizer set point and the supply air temperature is not met, the controller will increase the call for cooling to meet the supply air temperature and could engage mechanical cooling.

Temp./Enthalpy: The economizer will be locked out when: the outdoor air is < 40 F DB (adj.) or > 75 F DB (adj.) or > 55 F dew point (adj.) ; the unit is operating in dehumidification mode; or there is a call for heating.

ENERGY WHEEL SEQUENCE

Modulate Wheel: When economizer mode is enabled and there is a signal for cooling, the wheel VFD modulates wheel speed to maintain the supply air temperature set point.

ALARMS INDICATION: The controller will display alarms and have one digital output for remote indication of an alarm condition. Possible alarms include:

Building Management System: The controller will send all alarms to the BMS.

Dirty Wheel Alarm: The controller monitors pressure across the wheel and sends an alarm in the case of an increased pressure drop.

Wheel Rotation Alarm: The controller monitors wheel rotation, if the wheel does not rotate for a set period of time (adj.) an alarm will generate.

Supply and Exhaust Air Alarm: The controller monitors the proving switch on each blower and sends an alarm in the case of either blower proving switch not engaging.

DX Alarm: The controller monitors the refrigerant pressure. In the case of low refrigerant pressure the compressors will shut down until refrigerant pressure returns to normal values and the controller will send an alarm. In the case of high refrigerant pressure the compressors will shut down, requiring a manual reset and the controller will send a alarm.

Temperature Sensor Alarm: The controller sends an alarm in the case of a failed air temperature sensor.

Humidity Sensor Alarm: The controller sends an alarm in the case of a failed humidity sensor.

ACCESSORIES: The following accessories will be included with the unit to expand the functionality or usability of the controller.

BMS Interfacing: A BMS port or serial card is provided with the controller for field interfacing with a building management system. Each card is sent out with the default parameters, and the controls contractor must change the appropriate addresses to match the BMS settings.

DDC Remote Interface: Factory provided, field mounted interface panel that will be wired to the main controller for monitoring and remote adjustments of set points.

Phase and Brownout Protection: Factory mounted and wired component which monitors the main power coming into the unit. If a phase drops out, or if the incoming voltage exceeds the acceptable range, the component will turn off the unit to help protect the electrical systems.

Condensate Overflow Unit Shutdown: Factory mounted condensate overflow switch wired to the unit controller. The controller monitors the condensate overflow switch. If the water level in the drain pan reaches a certain level, the unit will shutdown and send an alarm.

Damper End Switch: Damper end switch will be provided to ensure the supply and exhaust fans do not enable until the dampers are proven open.

120V/24V Photoelectric Smoke Detector: Duct smoke detectors are shipped loose for field mounting and wiring in the supply and exhaust air ducts. Each duct smoke detector contains 2 normally open and 2 normally closed contacts for alarm notification. (To disable unit based off smoke detection smoke detectors contacts must be field wired between R and G)

Warranty Statement for Dedicated Outdoor Air Systems (DOAS)

Unit Warranty

Greenheck warrants the equipment to be free from defects in material and workmanship for a period of 18 months from ship date. Initial startup must be completed within six months of the shipment date, and a startup report must be submitted to Greenheck.

Energy Wheel Warranty

The energy recovery wheel is warranted to be free from defects in material and workmanship for a period of 5 years from the shipment date. This warranty applies to all parts and components in the energy recovery cassettes with the exception of the motor.

Heat Exchanger Extended Warranty

Greenheck warrants the stainless steel heat exchanger to be free from defects in material and workmanship for a period of 25 years from the shipment date.

Warranty Notes

Any component which proves defective during the warranty period will be repaired or replaced at Greenheck's sole option when returned to our factory, transportation prepaid. All warranties do not include labor costs associated with troubleshooting, removal, or installation. Greenheck will not be liable for any consequential, punitive, or incidental damages resulting from use, repair, or operation of any Greenheck product. These warranties are exclusive and are in lieu of all other warranties, whether written, oral, or implied, including the warranty of merchantability and the warranty of fitness for a particular purpose. No person (including any agent or salesperson) has authority to expand Seller's obligation beyond the terms of this warranty, or to state that the performance of the product is other than that published by Seller.

As a result of our commitment to continuous improvement, Greenheck reserves the right to change specifications without notice.

GKD Roof Curb

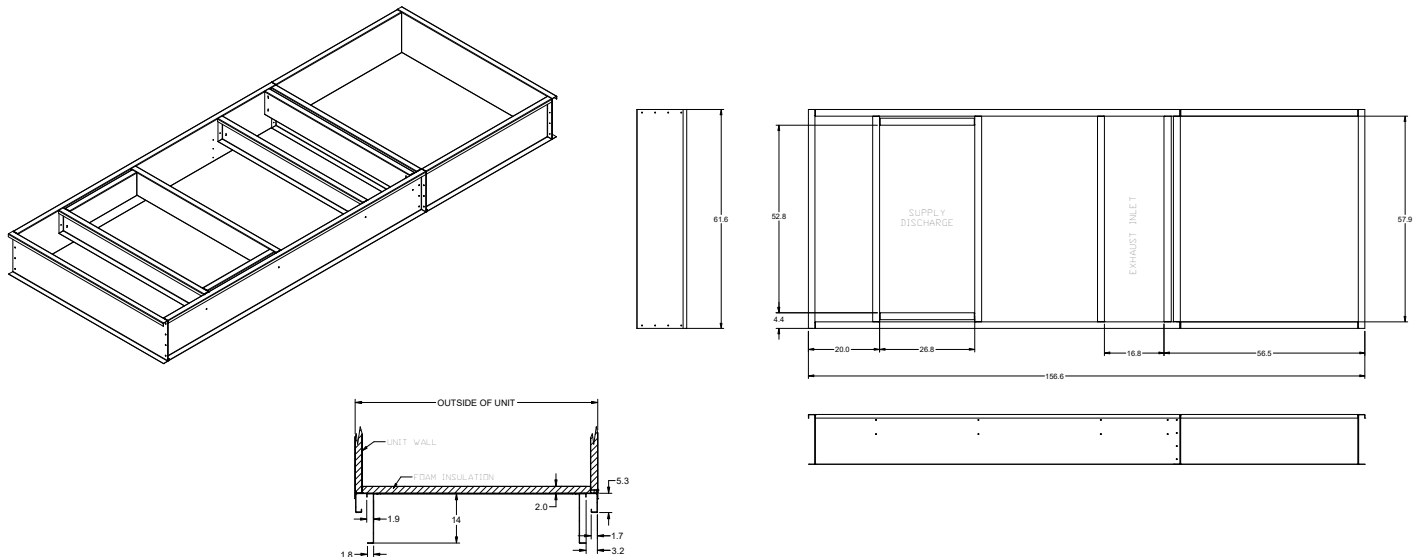
Model: GKD-61.6/156.6-G14

Curb Height (in.)	Curb Length (in.)	Curb Width (in.)	Material	Finish Type	Duct Adapter	Curb Weight (lb)
14	156.6	61.6	Galvanized	Galvanized	Yes	270

Standard Construction Features:

All dimensions shown are actual and in units of in.'s
If unit is selected with side or end discharge/return, there will not be bottom connections supplied with the curb.
14 gauge galvanized steel (perimeter channels).
14 gauge galvanized steel (interior channels).
14 gauge galvanized steel (perimeter channels).
14 gauge galvanized steel (interior channels).
Ships knocked down for field assembly.
Curb insulation to be provided by others.

Curb Detail



RVE-85-52D-20I-J-A2

Unit Performance

Design Conditions						
Elevation (ft)	Summer		Winter DB (F)	Supply (CFM)	Outdoor Air (CFM)	Exhaust Air (CFM)
	DB (F)	WB (F)				
23	88.9	75.0	3.5	6,370	6,370	5,775

Unit Specifications						
Qty	Weight (lb)	Cooling Type	Heating Type	Unit Installation	Unit ETL Listing	Furnace ETL Listing
1	4,685 (+/- 5%)	Packaged DX	Indirect Gas	Outdoor	UL\cUL 1995/60335-2-40	ANSI Z83.8 / CSA 2.6

Configuration			
Outdoor Air		Exhaust Air	
Intake	Discharge	Intake	Discharge
End	Bottom	Bottom	Side

ASHRAE 90.1 Compliance			
	ASHRAE 90.1 Min. Efficiency	Calculated Efficiency	Compliance
ISMRE2 (ASHRAE 90.1-2022)	5.0	7.2	✓
Enthalpy Recovery Ratio (%)	50	59.7	✓

Energy Recovery Performance									
Design Condition	Temperature (F)								Capacity Reduction (BTU/h)
	Outdoor Air		Supply Air		Return Air		Exhaust Air		
	DB	WB	DB	WB	DB	WB/RH	DB	WB	
Summer	88.9	75.0	80.3	68.0	75.0	62.5/50	84.3	71.1	177,723.0
Winter	3.5	1.7	43.4	37.5	72.0	55.8/35	25.4	24.2	274,496.0

Cooling Specifications							
Type	Total Capacity (MBH)	Sensible Capacity (MBH)	Lead Compressor Type	Coil (DB/WB)		Reheat	
				EAT (F)	LAT (F)	Capacity (MBH)	LAT (F)
Packaged DX	281.8	179.8	Inverter Scroll	80.3 / 68.0	53.4 / 53.4	200.6	82.7

Heating Specifications								
Type	Gas Type	Input (MBH)	Output (MBH)	Temperature Rise		Turndown	Performance	
				Min (F)	Max (F)		EAT (F)	LAT (F)
Indirect Gas	Natural	300.0	243.0	3.0	35.0	12:1	43.4	78.7

Motor Specifications						
Motor	Qty	Operating Power (hp)	Size (hp)	Enclosure	Efficiency	RPM
Supply	1	7.39	7-1/2	ODP	PE	1770
Exhaust	1	4.2	5	ODP	PE	1750

Electrical Specifications					
Power Supply	Rating (V/C/P)	MCA (A)	MOP (A)	FLA (A)	Fan Power (W/CFM)*
Unit	208/60/3	129.9	150.0	120.3	1.356

*Fan Power (W/CFM) = (Supply BHP + Exhaust BHP) / Supply CFM

Construction Features And Accessories

Unit	
Unit Installation - Outdoor	Std
Unit Construction - Double Wall	Std
Insulation - 2 inch 2.4# R13 foam	Std
Corrosion Resistant Fasteners	Std
Hinged Access	Std
Factory Wired Non-Fused Disconnect Switch	Std
Direct Drive Plenum Blower & Motor Assemblies	Std
Factory Wired VFDs	Std
Unit Finish - Permatector, Concrete Gray (RAL 7023)	X
Stainless Steel Condensate Drain Pan and Connection	Std
Condensate Drain Trap	Std
Short Circuit Current - 5 kA	Std
Energy Recovery Device - Polymer Wheel w/ Silica Gel Desiccant	Std
Controls	
Unit Controls - Full Control	Std
Internally Mounted Control Center with 24 VAC control transformer(s) and control circuiting fusing	Std
BMS Protocol - BACNetMSTP	X
BMS Monitoring Points	
Supply Fan Control - CO2 - By Factory	X
Exhaust Fan Control - Space Static Pressure by Factory	X
Economizer Control - Temp./Enthalpy	X
Exhaust Fan Only Power	
Web-Based User Interface	Std
Energy Wheel Economizer Control - Modulating Wheel, OA Temp Setpoint w/VFD Wheel	X
Energy Wheel Rotation Sensor	Std
Damper Control - 100% OA-No Recirculation	X
Unoccupied Recirc Mode	
Control Accessories	
Remote Display - w/150 ft cord	X
Dirty Filter Sensor(s)	
Airflow Monitor	
Room Thermostat - Space Temp and RH	X
Phase/Brownout Protection	Std
Economizer Fault Detection Diagnostics	

Accessories	
Frost Control Modulating Wheel - Modulating Wheel	X
Outdoor Air Damper - Low Leakage	X
Return Air Damper - Low Leakage	X
Roof Curb - GKD - 61.6/156.6-G14	X
Supply Air Filters - 2" Merv 13, 3-16x20x2, 6-20x20x2	X
Service Outlet - Factory mounted and wired	X
Piping Vestibule	
Service Lights	
Condensate Overflow Switch	X
Spare Filters	
Exhaust Discharge Gravity Backdraft Damper	X
ElectroFin Coil Coating	
Motor Shaft Grounding	
Bipolar Ionization	
Smoke Detector(s), Both - Shipped Loose	X
Barometric Relief Damper	
UV Lights	
Return Air Filters - 2" Merv 8, 4-16x25x2	Std
Outdoor Air Filters - 2" Merv 8, 4-16x25x2	Std
Furnace Control - 12:1 Modulating	X
Spare Energy Wheel Belt	
Spare Energy Wheel Segments	
Energy Wheel Bypass Damper	
Power Venting	Std
Hail Guards	
Warranty Options	
Unit Warranty - 18 Months (Std.)	Std
Energy Wheel Warranty - 5 Yrs Less Motor	Std
Compressor Warranty - 18 Months (Std.)	Std
Furnace HX Warranty - 25 Yrs.	Std

Standard Option	Std
Not Included	
Included	X

Notes	
Outdoor Air Damper supplied is low leakage, motorized VCD-23 (leakage rate of 3 CFM/ft ² @ 1 in. wg), Class 1A	
Return Air Damper supplied is low leakage, motorized VCD-23 (leakage rate of 3 CFM/ft ² @ 1 in. wg), Class 1A	
Smoke Detector(s) - sampling tube included sized for duct widths of 1-2 ft.	

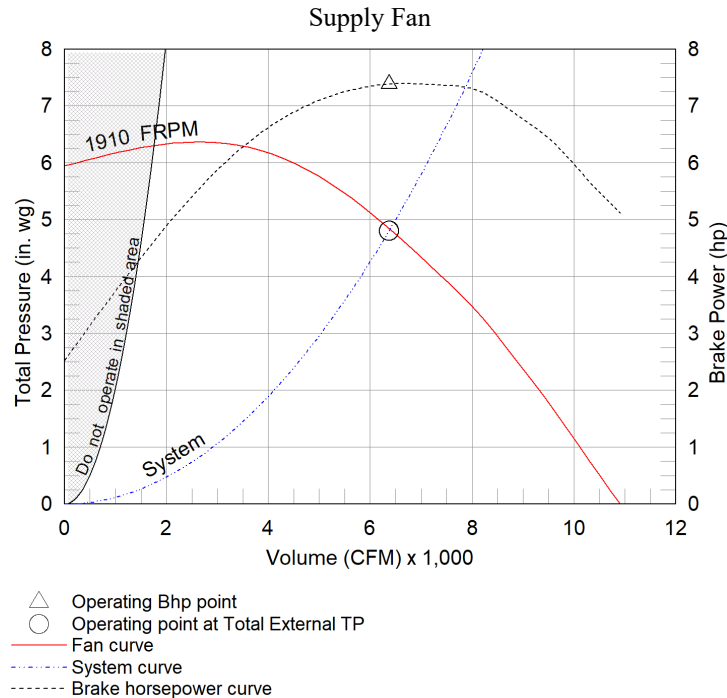
Supply Fan Charts And Performance

Supply Fan Performance									
Total Volume (CFM)	External SP (in. wg)	Total SP (in. wg)	RPM	Operating Power (hp)	Motor		Fan		
					Qty	Size (hp)	Qty	Type	Drive-Type
6,370	1.5	4.803	1910	7.39	1	7-1/2	1	Plenum	Direct

Pressure Drop (in. wg)							
Weatherhood	Filter	Damper	Cooling	Heating	External	Energy Wheel	Total
0.22	0.2	0.07	0.383	0.759	1.5	1.62	4.803

Sound Performance in Accordance with AMCA									
Sound Power by Octave Band								Lwa	dBA
62.5	125	250	500	1000	2000	4000	8000		
91	87	98	85	82	76	76	73	91	80
								Sones	
								33	

*Energy Wheel pressure drop shown in above table also accounts for pressure drop across MERV8 OA filter



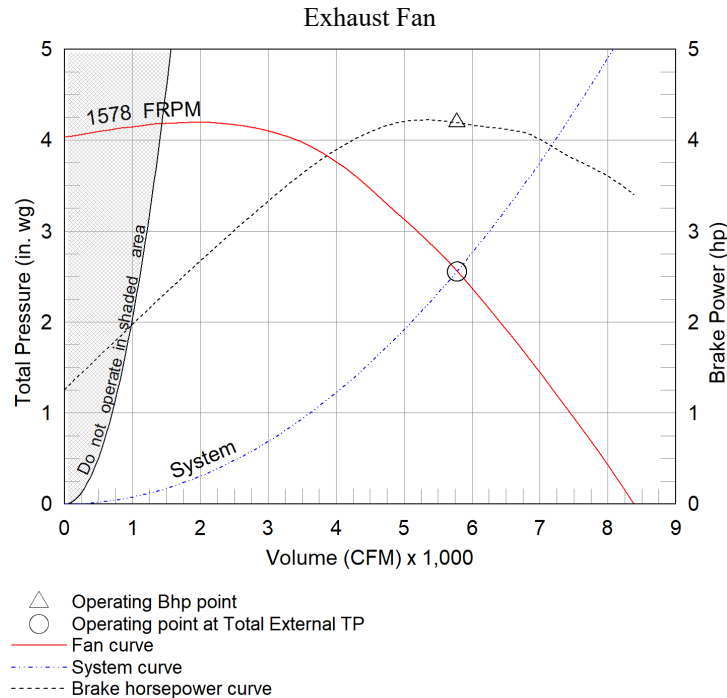
Exhaust Fan Charts And Performance

Exhaust Fan Performance										
Mode	Total Volume (CFM)	External SP (in. wg)	Total SP (in. wg)	RPM	Operating Power (hp)	Motor		Fan		
						Qty	Size (hp)	Qty	Type	Drive-Type
Normal	5,775	1.25	2.557	1578	4.2	1	5	1	Plenum	Direct

Pressure Drop (in. wg)								
Mode	Weatherhood	Filter	Damper	Cooling	Heating	External	Energy Wheel	Total
Normal	0.08	-	-	-	-	1.25	1.22	2.557

Sound Performance in Accordance with AMCA											
Mode	Sound Power by Octave Band								Lwa	dBA	Sones
	62.5	125	250	500	1000	2000	4000	8000			
Normal	92	84	87	75	69	67	66	65	81	69	19

*Energy Wheel pressure drop shown in above table also accounts for pressure drop across MERV8 return air filter

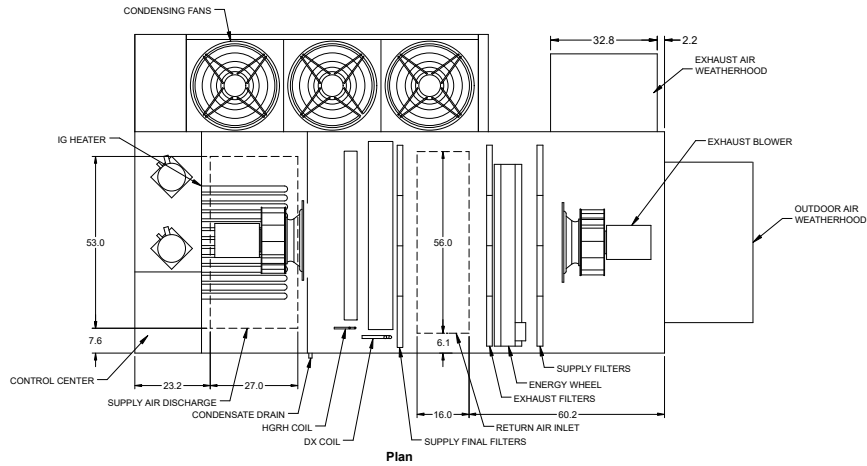


Radiated Sound

Position A

Dimensional Overview

Position D



Position B

Position C

"E" is the
Top Plane

Supply Air Flow Nominal, Largest Tonnage Condensing Section Available, PDX units only

Radiated Sound Levels										
Plane	Octave Bands (Lw)								Plane Lw	Plane LwA
	1	2	3	4	5	6	7	8		
A	80	87	82	80	77	72	68	62	90	82
B	74	80	73	67	64	59	55	48	82	71
C	82	77	73	69	67	61	54	46	84	72
D	71	80	74	72	69	63	58	52	82	74
E	77	86	82	78	76	71	66	61	89	81
Total	86	91	86	83	80	75	71	65	94	85

AMCA 320-07 - Laboratory Methods of Sound Testing of Fans Using Sound Intensity	
Tests conducted in accordance with this standard.	
Free field measurement plane created 1 foot from unit on all sides and top.	
Sound Intensity measured in Watts/m ² .	
Sound data converted to Sound Power (Lw) for the chart above.	
A-Weighted Sound Power was determined using AMCA Standard 301-90 Clause 9.1.	
Plane E sound data was measured above the top plane of the unit.	

Cooling Performance

Cooling Specifications									
Nominal Tonnage	Entering Air (F)		Leaving Air (F)		Capacity (MBH)		Reheat		Condensing Ambient Temp (F)
	DB	WB	DB	WB	Total	Sensible	Capacity (MBH)	LAT (F)	
20.0	80.3	68.0	53.4	53.4	281.8	179.8	200.6	82.7	88.9

Coil Information									
PDX Coil Model	Fins Per Inch	Rows Deep	Face Vel. (ft/min)	Coil PD (in. wg)	Refrigerant	Refrig. Velocity (ft/min)	Face Area (ft2)	Suction Temp (F)	
DR38S06H14-58x56-LH	14	6	282	0.383	R-454B	1,280	22.6	49.8	

Compressor Details					
Lead Compressor Type	Compressor Qty	Compressor RLA/MRC (A)		Compressor LRA (A)	
		Comp. #1	Comp. #2	Comp. #1	Comp. #2
Inverter Scroll	2	38.4	33.3	NA	255

A2L Installation Requirement - UL 60335-2-40		
Largest Circuit Charge	Minimum Circulation Airflow	Minimum Total Conditioned Room Area
48.6lb / 22.06kg	1,318 CFM	730 ft2

Local codes and standards may have requirements regarding the installation of A2L refrigerants in addition to manufacturing instructions provided for listed and labeled equipment.

Unit Details
Refrigerant charges provided by the factory are approximate and may require adjustment in the field
Hermetic scroll type compressors
Compressors mounted on neoprene vibration isolation
Crankcase heater on staged compressor
Electronic expansion valve on lead circuit, thermostatic expansion valve on staged circuit
Stainless steel double sloped drain pan
Moisture-indicating sight glass
Service/charging valves
Refrigerant high pressure switch (manual reset)
Liquid-Line filter drier
Leak detection sensors
Multiple low sound condensing fans with Lead ECM condensing fan for modulating head pressure control
Inverter scroll compressor
Refrigerant low pressure switch (auto reset)
Unit cannot be mounted in an enclosed space.

Important Notes:
Capacity is based on incoming voltage selected. If incoming power varies it may affect the capacity of your selection.

Heating Performance

Heating Specifications								
Type	Gas Type	Input (MBH)	Output (MBH)	Temperature Rise		Turndown	Performance	
				Min (F)	Max (F)		EAT (F)	LAT (F)
Indirect Gas	Natural	300.0	243.0	3.0	35.0	12:1	43.4	78.7

Unit Details
ANSI standard Z83.8 and CSA 2.6
High Thermal efficiency
Direct spark ignition
3/4" Gas Connection
At least 6 in. wg of natural gas pressure (14 in. wg for LP) is required at the units gas connection in order to achieve maximum performance
Power Venting
24 Volt Control Power
Stainless Steel heat exchange tubes
Unit controller maximum allowable supply discharge air set point is 100F (37.8C)
Discharge temperature assumes proper energy wheel operation and maintenance.

Energy Recovery Summer Performance

Outdoor Air		Supply Air	
Dry Bulb (F)	88.9	Dry Bulb (F)	80.3
Wet Bulb (F)	75.0	Wet Bulb (F)	68.0
Specific Humidity (gr/lb)	109	Specific Humidity (gr/lb)	83
Enthalpy (BTU/lb)	38.5	Enthalpy (BTU/lb)	32.3
Exhaust Air		Return Air	
Dry Bulb (F)	84.3	Dry Bulb (F)	75.0
Wet Bulb (F)	71.1	Rel. Humidity (%)	50
Specific Humidity (gr/lb)	92	Specific Humidity (gr/lb)	65
Enthalpy (BTU/lb)	34.7	Enthalpy (BTU/lb)	28.1

Design Air Flow Conditions

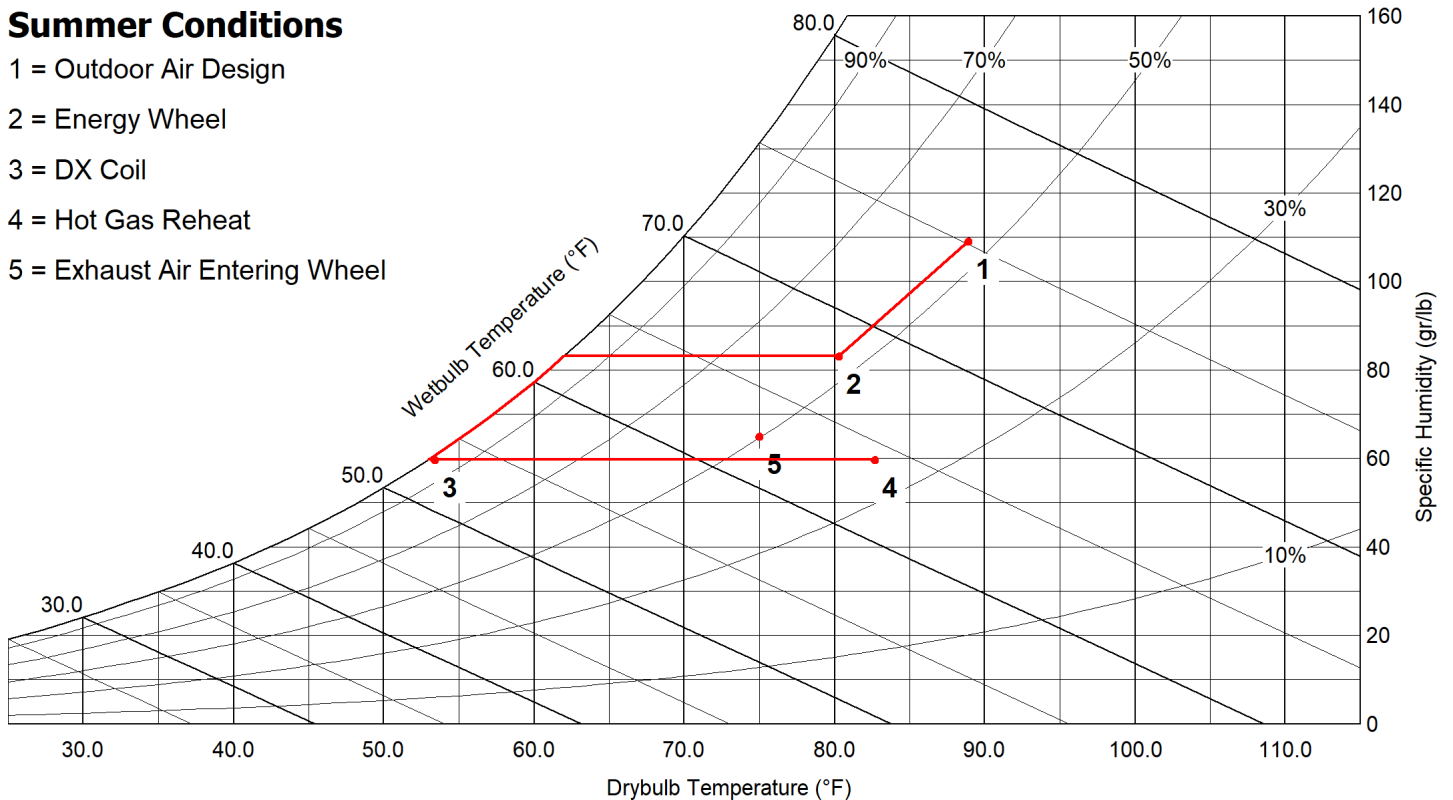
OA Volume (CFM)	ASHRAE 90.1 OA Enthalpy Recovery Ratio	EA Volume (CFM)	EA Wheel Effectiveness
6,370	59.7	5,775	63.9

Outdoor Air Cooling Reduction

OA Load w/o Energy Recovery		OA Load with Energy Recovery		Equipment Reduction (tons)
(BTU/h)	(tons)	(BTU/h)	(tons)	
298,116.0	24.84	120,393.0	10.03	14.81

Summer Conditions

- 1 = Outdoor Air Design
- 2 = Energy Wheel
- 3 = DX Coil
- 4 = Hot Gas Reheat
- 5 = Exhaust Air Entering Wheel



Energy Recovery Winter Performance w/out Preheater

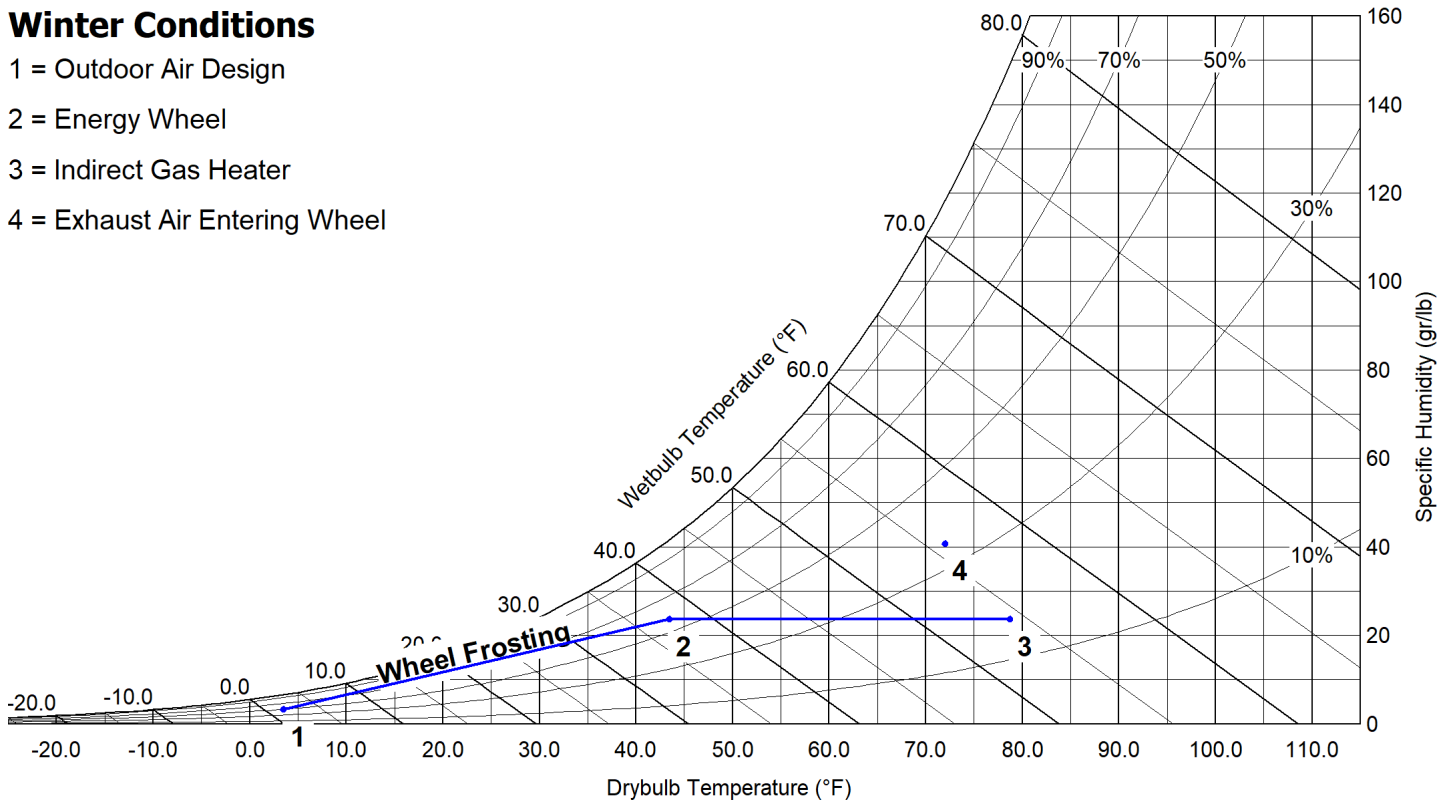
Outdoor Air		Supply Air	
Dry Bulb (F)	3.5	Dry Bulb (F)	43.4
Wet Bulb (F)	1.7	Wet Bulb (F)	37.5
Specific Humidity (gr/lb)	3	Specific Humidity (gr/lb)	24
Enthalpy (BTU/lb)	1.3	Enthalpy (BTU/lb)	14.1
Exhaust Air		Return Air	
Dry Bulb (F)	25.4	Dry Bulb (F)	72.0
Wet Bulb (F)	24.2	Rel. Humidity (%)	35
Specific Humidity (gr/lb)	17	Specific Humidity (gr/lb)	41
Enthalpy (BTU/lb)	8.7	Enthalpy (BTU/lb)	23.7

Design Air Flow Conditions			
OA Volume (CFM)	ASHRAE 90.1 OA Enthalpy Recovery Ratio	EA Volume (CFM)	EA Wheel Effectiveness
6,370	57.1	5,775	66.3

Outdoor Air Heating Reduction			
OA Load w/o Energy Recovery (BTU/h)	OA Load with Energy Recovery (BTU/h)	Equipment Reduction (BTU/h)	Sensible Effectiveness (%)
471,253.0	196,757.0	274,496.0	67.4

Winter Conditions

- 1 = Outdoor Air Design
- 2 = Energy Wheel
- 3 = Indirect Gas Heater
- 4 = Exhaust Air Entering Wheel



AHRI Performance Ratings

Energy Recovery Performance Rating in accordance with AHRI Standard 1060 (I-P)

Rated Airflow (SCFM)		Net Supply Airflow (SCFM)	EATR (%)	OACF	Pressure Drop (in. wg)		Purge Angle (degrees)
Leaving Supply	Entering Exhaust				Supply	Exhaust	
5,804	5934	6529	2.4	1.01	0.94	0.84	0

Thermal Effectiveness Ratings

Enthalpy Recovery		Sensible Effectiveness		Latent Effectiveness		Total Effectiveness	
Summer	Winter	Summer	Winter	Summer	Winter	Summer	Winter
59.7	57.1	66.7	67.4	62.5	63.1	63.9	66.3

Note(s)

Summer Design Conditions:

Certified in accordance with the AHRI ERV Certification Program, which is based on AHRI Standard 1060. Certified units may be found in the AHRI Directory at www.ahridirectory.org.

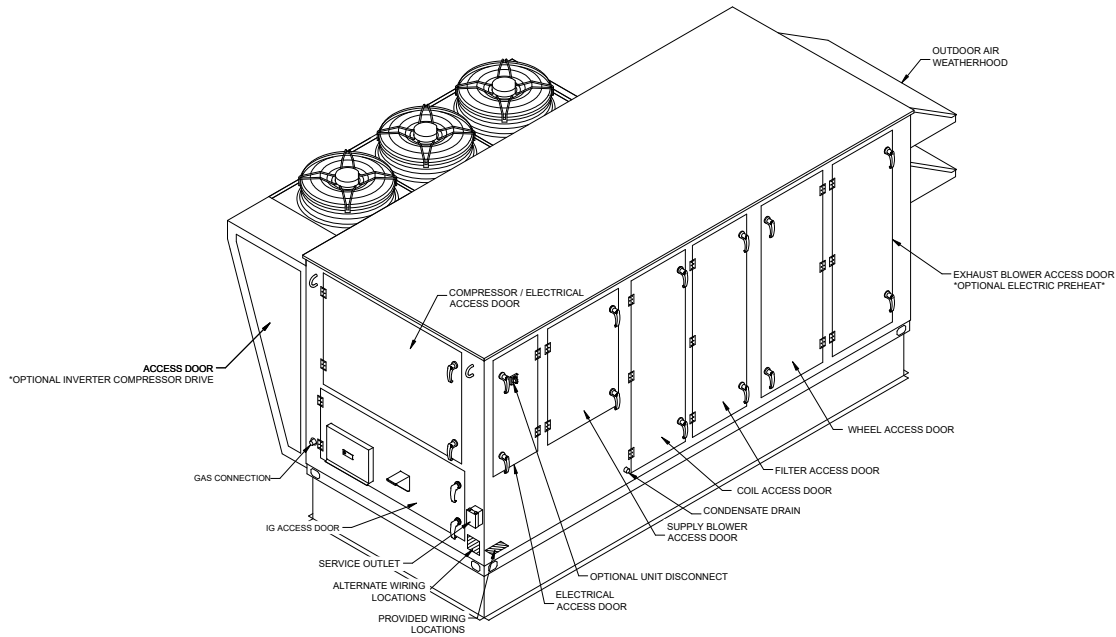


Winter Design Conditions:

Application Rating is outside the scope of the AHRI ERV certification Program but is rated in accordance with AHRI Standard 1060.

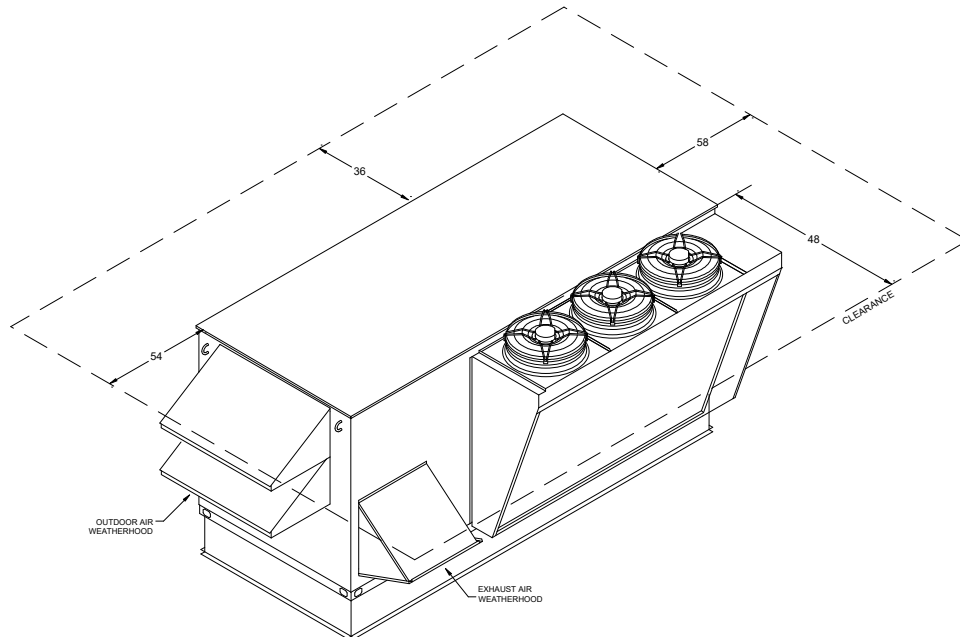
Isometric Drawings

Component Layout



Back Right Isometric

Service Clearances

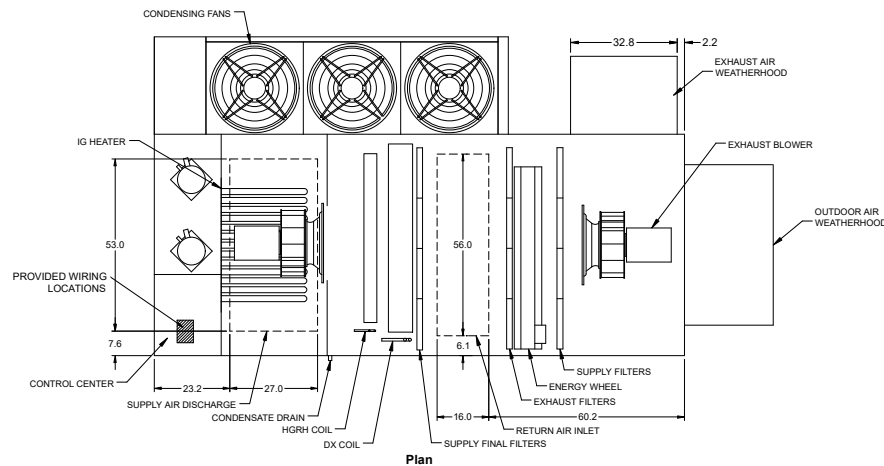
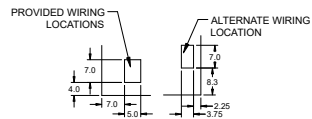


Front Left Isometric

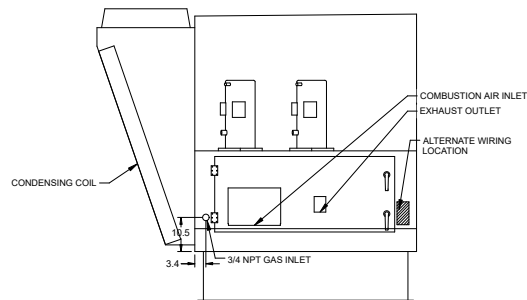
Overview Drawings

Dimensional Overview

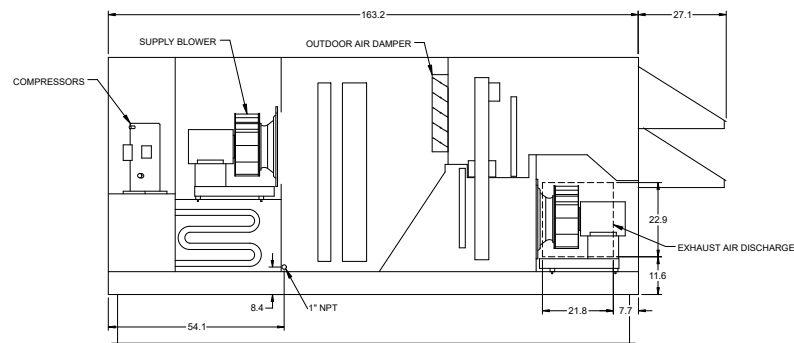
Electrical Connections



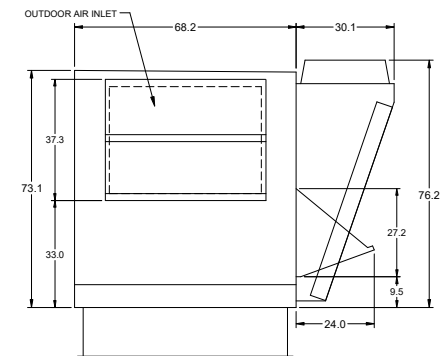
Plan



Left End

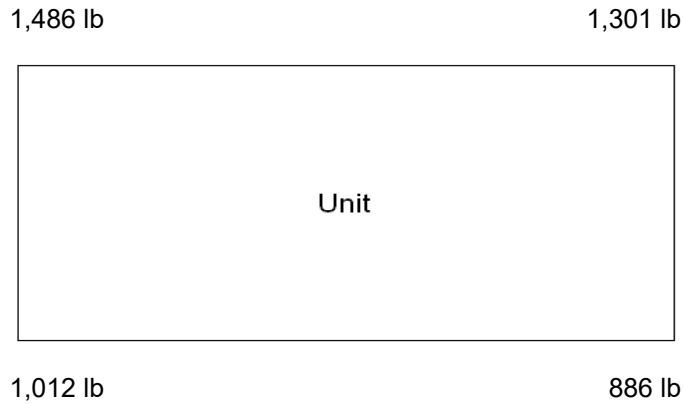


Elevation



Right End

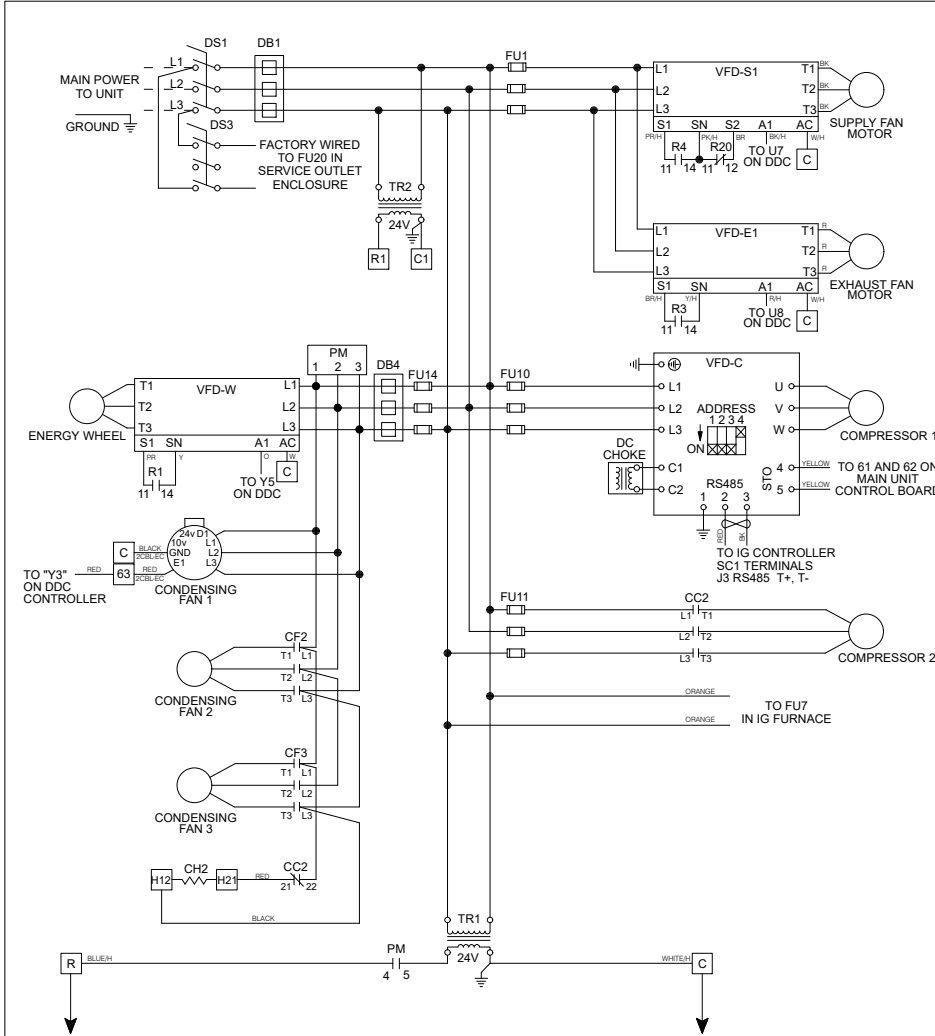
Unit Corner Weights



Note

Estimated corner weights are shown looking down on unit and the outside air intake will be on the right. Weights are applied at the base of the unit. Images not drawn to scale.

Wiring Diagram



Wiring Diagram Code:

G23N2E1BF02G25X00HJ23H0630BH31

CAUTION

UNIT SHALL BE GROUNDED IN ACCORDANCE WITH N.E.C.
POWER MUST BE OFF WHILE SERVICING.

NOTES

USE COPPER CONDUCTORS ONLY
60° C FOR TERMINALS RATED LESS THAN 100 AMPS.
75° C FOR TERMINALS RATED 100 AMPS OR MORE.
FIELD CONTROL WIRING RESISTANCE SHOULD NOT EXCEED 0.75 OHM.

FIELD WIRED - - - - -
FACTORY SUPPLIED AND WIRED _____

WIRE COLOR CODE

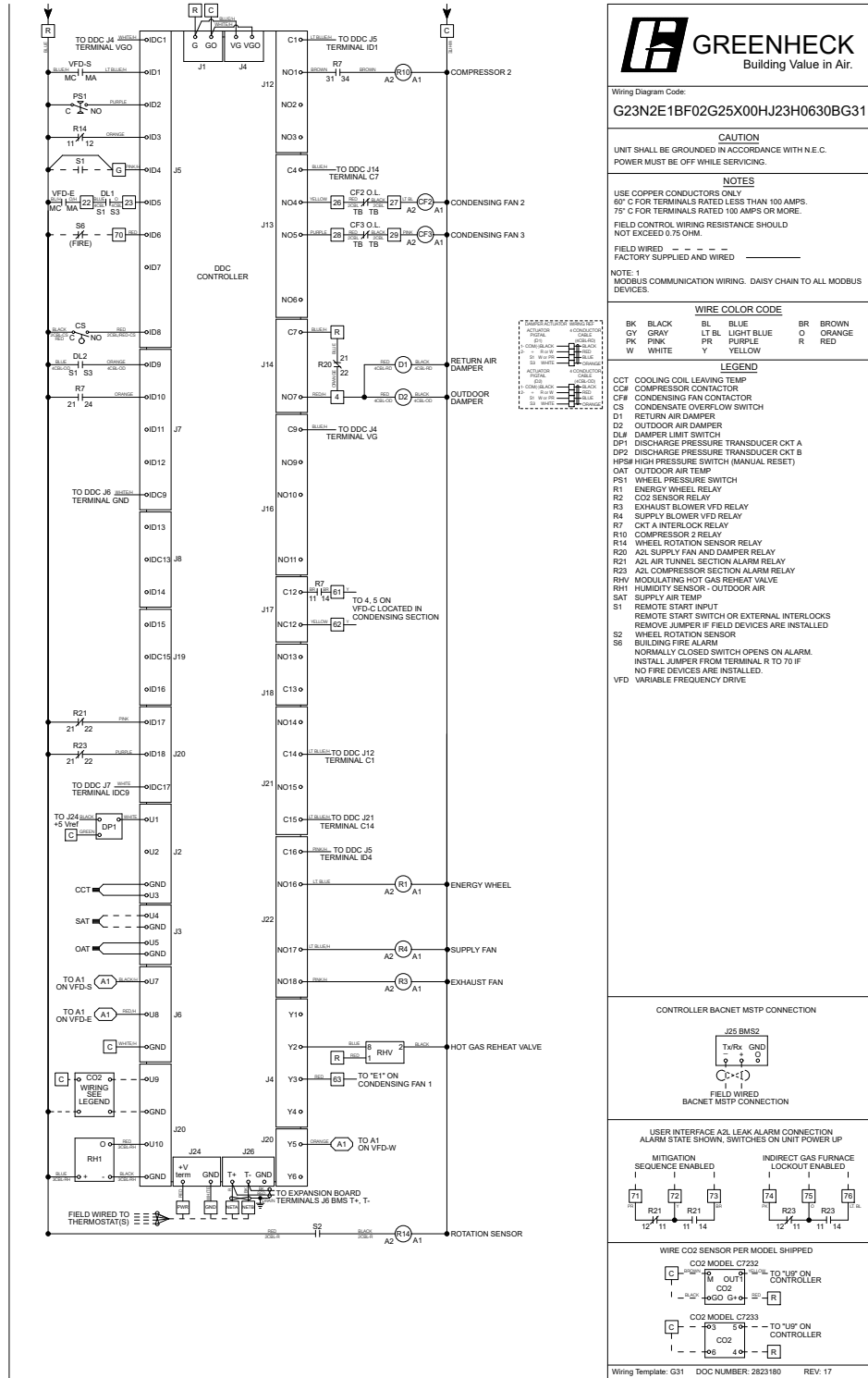
BK	BLACK	BL	BLUE	BR	BROWN
GY	GRAY	LT BL	LIGHT BLUE	O	ORANGE
PK	PINK	PR	PURPLE	R	RED
W	WHITE	Y	YELLOW		

LEGEND

CC#	COMPRESSOR CONTACTOR
CF#	CONDENSING FAN CONTACTOR
CH#	COMPRESSOR SUMP HEATER
DB#	POWER DISTRIBUTION BLOCK
DS	DISCONNECT SWITCH
FU#	FUSES
PM	PHASE VOLTAGE MONITOR
R1	ENERGY WHEEL RELAY
R3	EXHAUST BLOWER VFD RELAY
R4	SUPPLY BLOWER VFD RELAY
R20	A2L SUPPLY FAN AND DAMPER RELAY
TR#	TRANSFORMER
VFD	VARIABLE FREQUENCY DRIVE

Wiring Template: H31 DOC NUMBER: 2823180 REV: 17

Wiring Diagram 2



Wiring Diagram 3



GREENHECK
Building Value in Air.

Wiring Diagram Code:

G23N2E1BF02G25X00HJ23H0630BL31

CAUTION

UNIT SHALL BE GROUNDED IN ACCORDANCE WITH N.E.C.
POWER MUST BE OFF WHILE SERVICING.

NOTES

USE COPPER CONDUCTORS ONLY
60° C FOR TERMINALS RATED LESS THAN 100 AMPS.
75° C FOR TERMINALS RATED 100 AMPS OR MORE.
FIELD CONTROL WIRING RESISTANCE SHOULD NOT EXCEED 0.75 OHM.
FIELD WIRED - - - - -
FACTORY SUPPLIED AND WIRED _____

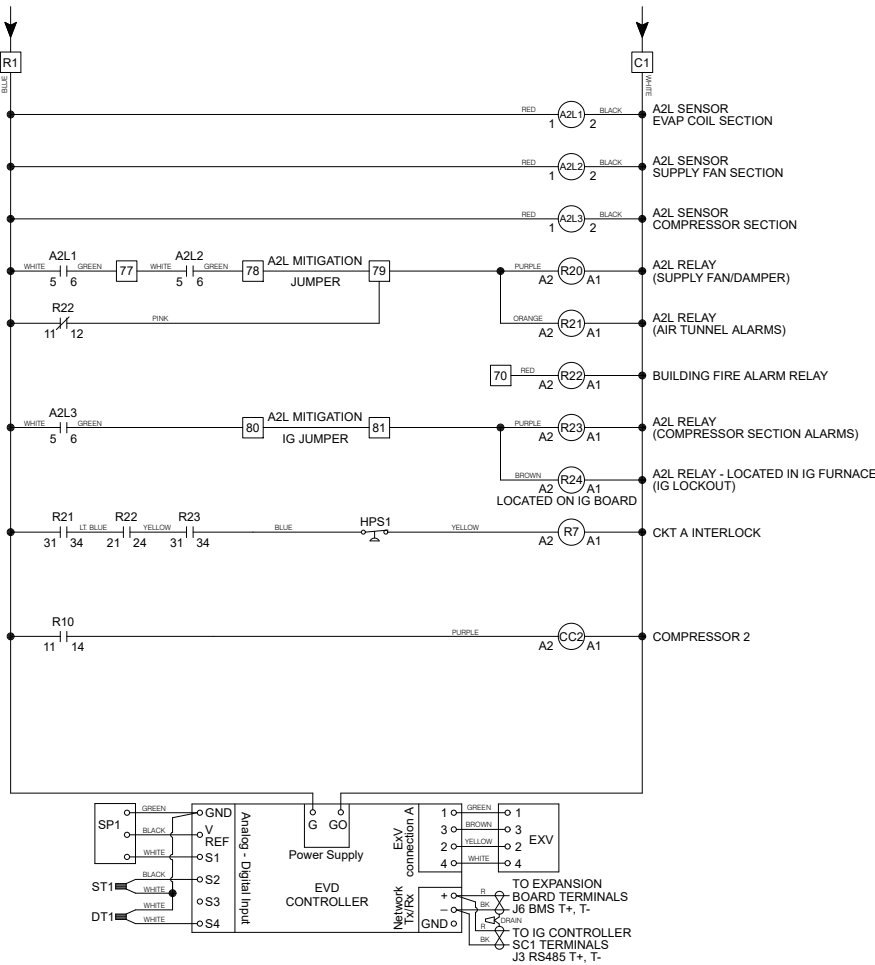
WIRE COLOR CODE

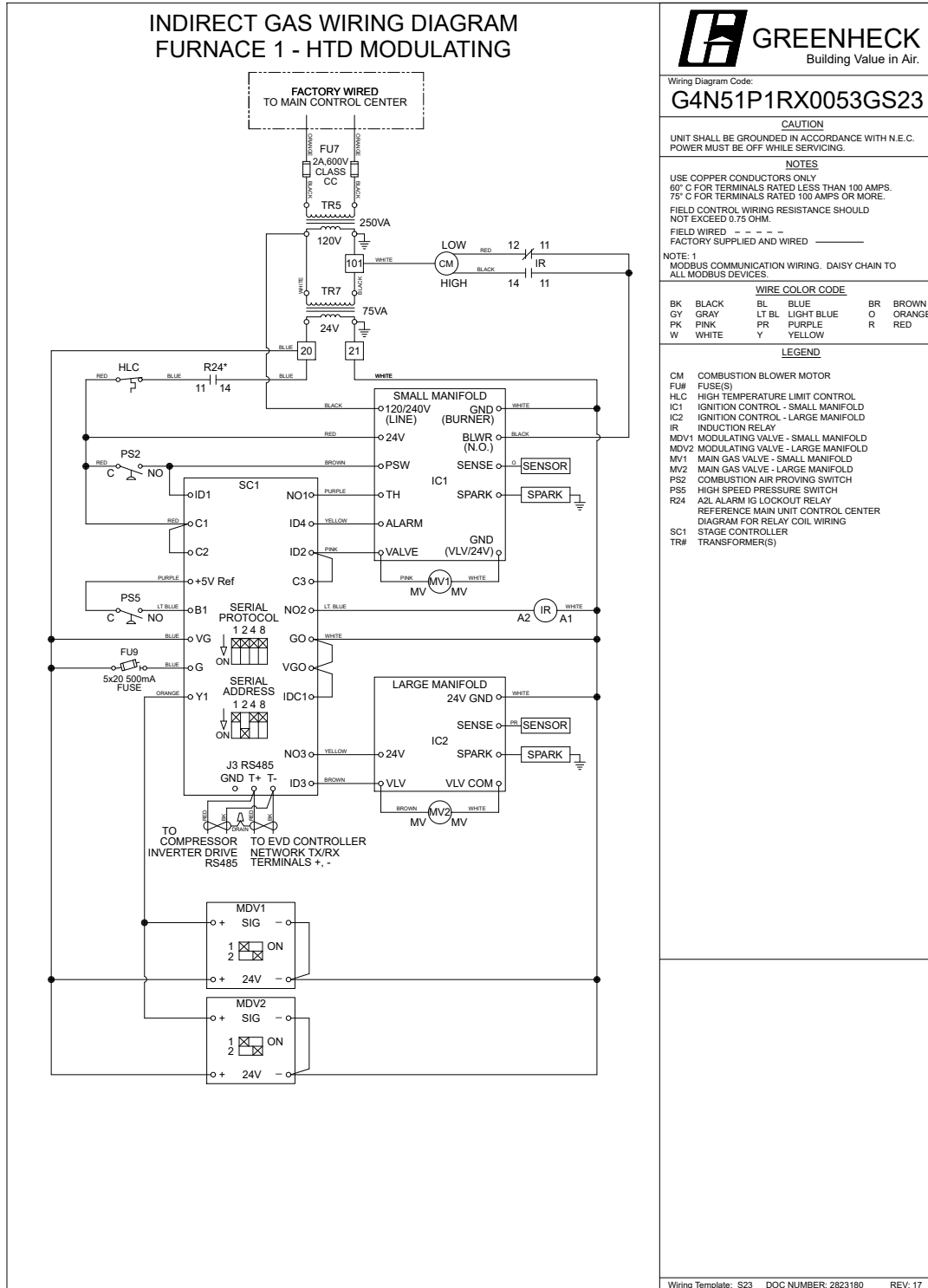
BK	BLACK	BL	BLUE	BR	BROWN
GY	GRAY	LT BL	LIGHT BLUE	O	ORANGE
PK	PINK	PR	PURPLE	R	RED
W	WHITE	Y	YELLOW		

LEGEND

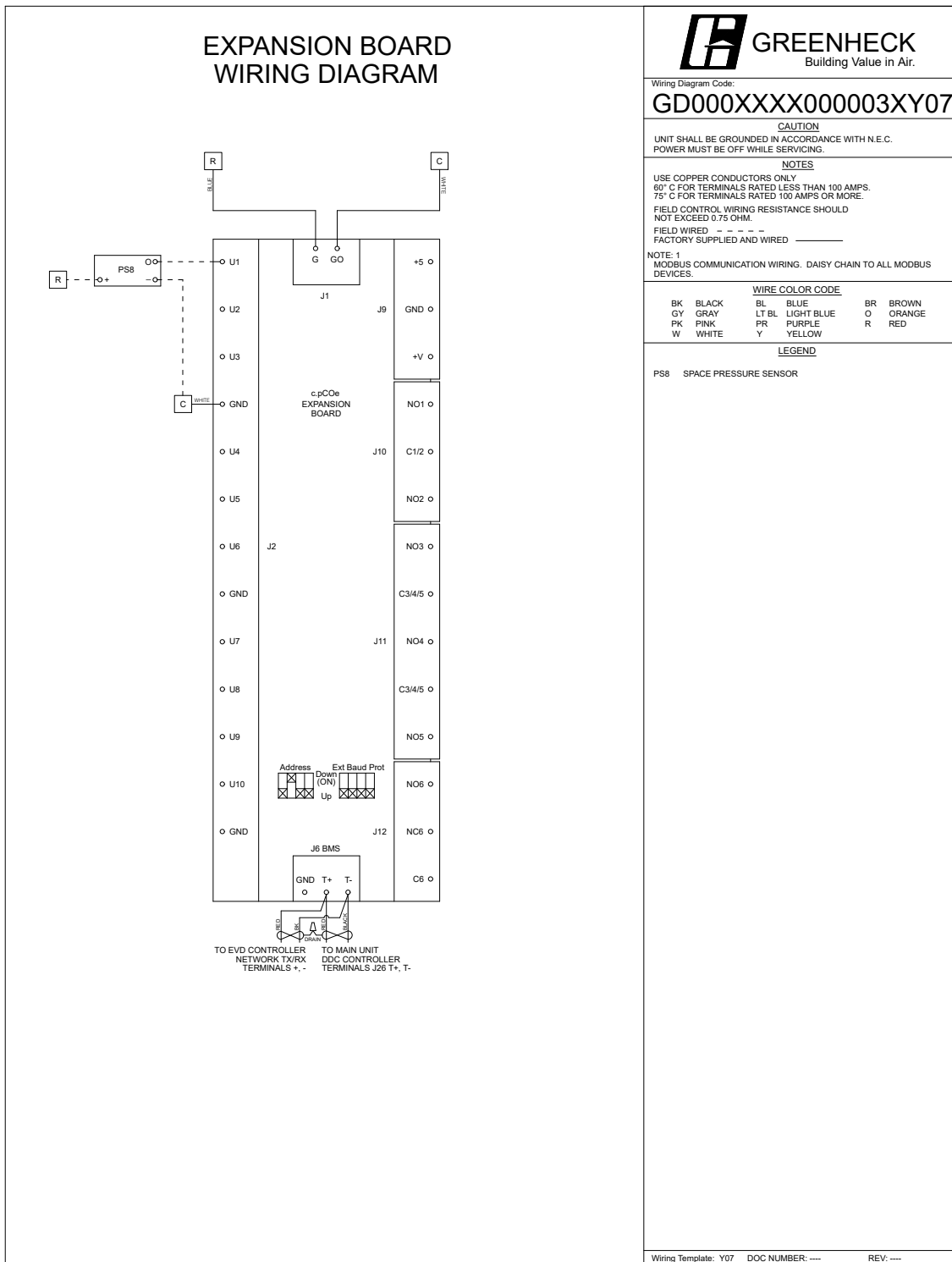
A2L# A2L REFRIGERANT LEAK SENSOR
CC# COMPRESSOR CONTACTOR
DT1 DISCHARGE LINE TEMP CKT A
EXV ELECTRONIC EXPANSION VALVE
R7 CKT A INTERLOCK RELAY
R10 COMPRESSOR 2 RELAY
R20 A2L SUPPLY FAN AND DAMPER RELAY
R21 A2L AIR TUNNEL SECTION ALARM RELAY
R22 BUILDING FIRE ALARM RELAY
R23 A2L COMPRESSOR SECTION ALARM RELAY
R24 A2L ALARM IG LOCKOUT RELAY
RELAY LOCATED ON IG FURNACE CONTROL CENTER BOARD
SP1 SUCTION PRESSURE TRANSDUCER CKT A
ST1 SUCTION LINE TEMP CKT A
VFD VARIABLE FREQUENCY DRIVE

Wiring Template: L31 DOC NUMBER: 2823180 REV: 17





Monitoring Points Wiring Diagram



Factory Controller Sequence of Operation

FACTORY CONTROLLER: Controller shall be provided with required sensors and programming for rooftop unit. Controller shall be factory programmed, mounted and tested. Controller shall have a LCD readout for changing set points and monitoring unit operation.

UNIT START COMMAND (Unit will be enabled to start once a jumper is placed between R to G):

- Factory mounted and wired outdoor air damper actuator is powered
- Return/Exhaust air damper actuator is powered.
- Exhaust fan starts after a (adj.) delay.
- Supply fan starts after a (adj.) delay.
- Tempering options and energy wheel option to function as described below.

UNIT STOP COMMAND (OR DE-ENERGIZED):

- Supply fan, exhaust fan, energy wheel and tempering options de-energized.
- Outdoor air damper actuator is spring return close.
- Return air damper is spring return close.

OCCUPIED/UNOCCUPIED MODES: Shall be based on a 7-day time clock internal to the controller. The schedule shall be set by the end user. When a user initiates an override input, the controller will switch from unoccupied to occupied mode. The controller will return to the scheduled occupied/unoccupied mode after the override time has expired. If internal time clock is disabled, a remote contact or a BMS can control the occupied/unoccupied mode.

Occupied Mode:

- Damper control per below.
- Energy wheel control per below.
- Exhaust fan ON.
- Supply fan ON.
- Heating per below.
- Cooling per below.

Unoccupied Mode (Unit Off): Unit remains off when in unoccupied mode.

- Supply fan OFF
- Exhaust fan OFF
- Tempering OFF
- Outdoor air damper closed.
- Return damper closed.

MORNING WARMUP/COOL DOWN: Prior to occupancy, the unit will run using the warmup or cool down sequence until the occupied set point is achieved. The heating or cooling mode must not be locked out and the space temperature is below or above set point by the unoccupied hysteresis (adj.) (This Sequence must be field configured.)

SUPPLY BLOWER SEQUENCE: The supply blower is provided with a factory mounted variable frequency drive. The supply blower speed will be controlled with the following sequence. Minimum supply fan turndown is 50% of the design maximum operation.

CO2 Sensor: The supply blower speed is modulated based upon the signal from a CO2 Sensor (factory provided, field mounted and wired). The controller will modulate the supply fan based upon a comparison of the CO2 set point (adj.) to the actual CO2 levels reported from the sensor. **(Mechanical high static protection cutoffs must be installed by others to protect the system and equipment from over-pressurization. The manufacturer does not assume responsibility for this.)**

EXHAUST BLOWER SEQUENCE: The exhaust blower will operate at a constant speed set point (adj.) during operation.

Space Static Pressure by Factory: The exhaust blower is modulated based upon the signal from a building pressure sensor. The controller will modulate the exhaust fan based upon a comparison of the building static pressure set point (adj.) and the building static pressure level reported from the sensor.

COOLING SEQUENCE: The cooling is controlled to maintain the supply temperature set point. The mechanical cooling will be locked out when the outside air is < 55 F (adj.).

Packaged DX Cooling (Inverter Scroll): The controller will provide a modulating signal for cooling. From 0-50%, the inverter scroll will be controlled to maintain discharge temperature. From 50-100% the second stage will be on in combination with the inverter scroll compressor to maintain the discharge temperature. The electronic expansion valve will modulate to maintain 8F of superheat.

Modulating Hot Gas Reheat Sequence: During dehumidification the modulating HGRH is controlled to maintain the supply temperature set point.

Modulating Head Pressure Control: Lead condenser fan will have an EC motor and will modulate to maintain a head pressure set point.

REHEAT SEQUENCE: While the unit is in dehumidification mode the outdoor air will be reheated via Modulating Hot Gas Reheat for space neutral applications.

Modulating Hot Gas Reheat: The controller will modulate the hot gas reheat valve with a 0-10 V signal to maintain the supply temperature set point (adj.).

HEATING SEQUENCE: The heating is controlled to maintain the supply temperature set point. The heating will be locked out when the outside air is > 80 F (adj.).

Indirect Gas Furnace: The controller will modulate the indirect gas furnace to maintain the supply temperature set point (adj.).

TEMPERATURE CONTROL SEQUENCE: The unit will maintain the supply air discharge setpoint per the following. Adjustable locally or by BMS.

Room Reset: The controller will reset the supply air temperature set point to maintain the room temperature set point (adj.).

Space Setpoint Control: The supply setpoint will adjust between minimum (adj.) and maximum (adj.) limits, to satisfy the desired space temperature setpoint. Adjustable locally or by BMS.

BUILDING FREEZE PROTECTION: If the supply air temperature drops below 35 F (adj.) for 300s (adj.), the controller will de-energize the unit and activate the alarm output.

TEMPERATURE PROTECTION: The controller will enable the supply fan to modulate down to help the unit keep up with heating demand in the event of wheel failure or the unit operating outside design conditions. (This can be enabled under the manufacturer menu in the controller)

UNIT LEAK DETECTION AND MITIGATION: The unit will be equipped with refrigerant leak detection sensors. These sensors along with the following sequence of operation are required per UL60335-2-40.

Refrigerant Leak Detected In Air Tunnel: If a refrigerant leak is detected in the air tunnel, the supply fan will operate at minimum airflow requirement, outside air damper will be opened, powered exhaust fan will operate based configured sequence, energy recovery wheel will operate if commanded by normal unit sequencing, and compressors are disabled to reduce leakage rate. This operation is required in order to move stagnant refrigerant from within the unit, duct, and space ensuring proper dilution of the refrigerant. This operation is required even if the unit is called to be off. After leak detection is cleared, the unit will go back to normal operation. Exception to this operation is when the unit is receiving an active fire alarm signal at the unit controller. If unit controller is receiving a fire alarm input, the unit will not operate the leak mitigation supply fan sequence.

Dry alarm contacts available to allow the building (by others) to perform external mitigation actions when necessary. These by other external actions include opening of zone dampers in the ductwork, disabling duct mounted electric resistance heaters, and/or enabling additional mechanical ventilation if required per ASHRAE 15.

Refrigerant Leak Detected In Compressor Compartment: If a refrigerant leak is detected in the compressor compartment and the unit is configured with an indirect gas furnace, the furnace will be disabled while leak detection is active. After leak detection is cleared, the unit will go back to normal operation.

ENERGY WHEEL FROST CONTROL: Frost control for the energy wheel is enabled when frost is present on the wheel; based on the outside air temperature and the pressure drop across the wheel. If the outdoor air temperature is below 5 F adj. and the differential pressure across the wheel is about 1.5", adj. frost control will enable.

Wheel VFD (Modulate Wheel): When frosting is occurring, the VFD modulates the wheel down to a slow rotational speed to defrost wheel. Once either the pressure drop decreases below the pressure switch set point, or the outdoor air temperature increases about the temperature set point, the unit will resume normal operation.

ECONOMIZER SEQUENCE: When the application requires cooling, and the outdoor air conditions are suitable for free cooling, the controller will modulate the energy wheel speed to maintain the discharge temperature set point. If the energy wheel speed modulates to the economizer set point and the supply air temperature is not met, the controller will increase the call for cooling to meet the supply air temperature and could engage mechanical cooling.

Temp./Enthalpy: The economizer will be locked out when: the outdoor air is < 40 F DB (adj.) or > 75 F DB (adj.) or > 55 F dew point (adj.) ; the unit is operating in dehumidification mode; or there is a call for heating.

ENERGY WHEEL SEQUENCE

Modulate Wheel: When economizer mode is enabled and there is a signal for cooling, the wheel VFD modulates wheel speed to maintain the supply air temperature set point.

ALARMS INDICATION: The controller will display alarms and have one digital output for remote indication of an alarm condition. Possible alarms include:

Building Management System: The controller will send all alarms to the BMS.

Dirty Wheel Alarm: The controller monitors pressure across the wheel and sends an alarm in the case of an increased pressure drop.

Wheel Rotation Alarm: The controller monitors wheel rotation, if the wheel does not rotate for a set period of time (adj.) an alarm will generate.

Supply and Exhaust Air Alarm: The controller monitors the proving switch on each blower and sends an alarm in the case of either blower proving switch not engaging.

DX Alarm: The controller monitors the refrigerant pressure. In the case of low refrigerant pressure the compressors will shut down until refrigerant pressure returns to normal values and the controller will send an alarm. In the case of high refrigerant pressure the compressors will shut down, requiring a manual reset and the controller will send a alarm.

Temperature Sensor Alarm: The controller sends an alarm in the case of a failed air temperature sensor.

Humidity Sensor Alarm: The controller sends an alarm in the case of a failed humidity sensor.

ACCESSORIES: The following accessories will be included with the unit to expand the functionality or usability of the controller.

BMS Interfacing: A BMS port or serial card is provided with the controller for field interfacing with a building management system. Each card is sent out with the default parameters, and the controls contractor must change the appropriate addresses to match the BMS settings.

DDC Remote Interface: Factory provided, field mounted interface panel that will be wired to the main controller for monitoring and remote adjustments of set points.

Phase and Brownout Protection: Factory mounted and wired component which monitors the main power coming into the unit. If a phase drops out, or if the incoming voltage exceeds the acceptable range, the component will turn off the unit to help protect the electrical systems.

Condensate Overflow Unit Shutdown: Factory mounted condensate overflow switch wired to the unit controller. The controller monitors the condensate overflow switch. If the water level in the drain pan reaches a certain level, the unit will shutdown and send an alarm.

Damper End Switch: Damper end switched will be provided to ensure the supply and exhaust fans do not enable until the dampers are proven open.

120V/24V Photoelectric Smoke Detector: Duct smoke detectors are shipped loose for field mounting and wiring in the supply and exhaust air ducts. Each duct smoke detector contains 2 normally open and 2 normally closed contacts for alarm notification. (To disable unit based off smoke detection smoke detectors contacts must be field wired between R and G)

Warranty Statement for Dedicated Outdoor Air Systems (DOAS)

Unit Warranty

Greenheck warrants the equipment to be free from defects in material and workmanship for a period of 18 months from ship date. Initial startup must be completed within six months of the shipment date, and a startup report must be submitted to Greenheck.

Energy Wheel Warranty

The energy recovery wheel is warranted to be free from defects in material and workmanship for a period of 5 years from the shipment date. This warranty applies to all parts and components in the energy recovery cassettes with the exception of the motor.

Heat Exchanger Extended Warranty

Greenheck warrants the stainless steel heat exchanger to be free from defects in material and workmanship for a period of 25 years from the shipment date.

Warranty Notes

Any component which proves defective during the warranty period will be repaired or replaced at Greenheck's sole option when returned to our factory, transportation prepaid. All warranties do not include labor costs associated with troubleshooting, removal, or installation. Greenheck will not be liable for any consequential, punitive, or incidental damages resulting from use, repair, or operation of any Greenheck product. These warranties are exclusive and are in lieu of all other warranties, whether written, oral, or implied, including the warranty of merchantability and the warranty of fitness for a particular purpose. No person (including any agent or salesperson) has authority to expand Seller's obligation beyond the terms of this warranty, or to state that the performance of the product is other than that published by Seller.

As a result of our commitment to continuous improvement, Greenheck reserves the right to change specifications without notice.

GKD Roof Curb

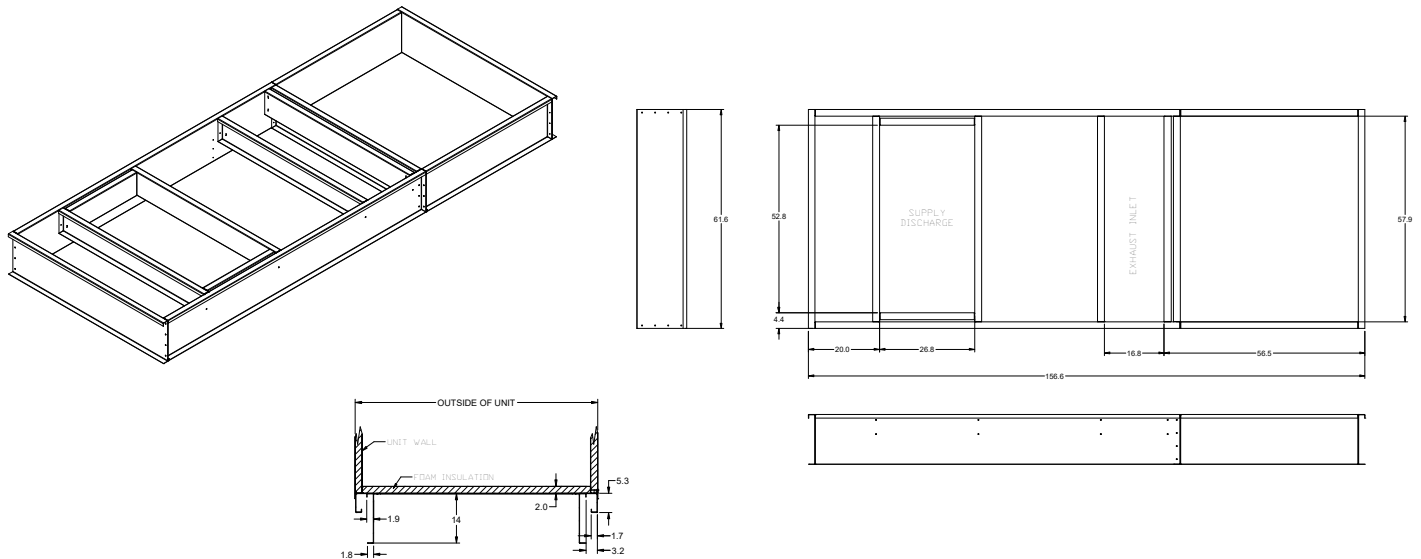
Model: GKD-61.6/156.6-G14

Curb Height (in.)	Curb Length (in.)	Curb Width (in.)	Material	Finish Type	Duct Adapter	Curb Weight (lb)
14	156.6	61.6	Galvanized	Galvanized	Yes	270

Standard Construction Features:

All dimensions shown are actual and in units of in.'s
If unit is selected with side or end discharge/return, there will not be bottom connections supplied with the curb.
14 gauge galvanized steel (perimeter channels).
14 gauge galvanized steel (interior channels).
14 gauge galvanized steel (perimeter channels).
14 gauge galvanized steel (interior channels).
Ships knocked down for field assembly.
Curb insulation to be provided by others.

Curb Detail





MERV 8 Efficiency	
Standard Capacity:	.21 Initial Resistance (In. W.G.)
High Capacity:	.19 Initial Resistance (In. W.G.)

DP®/DP Max

Extended Surface Pleated Filters



- Standard and high capacity MERV 8, MERV-A 8-A
- Low Initial Resistance for energy savings
- Extremely high dust-loading
- Quality engineered for consistent production
- Industry's largest Inventory
- Proudly made in the U.S.A.

DP®/DP Max

Extended Surface Pleated Filters

**Exceeds ASHRAE Standard 52.2
air cleaning specifications for
filters installed upstream of
cooling coils.**

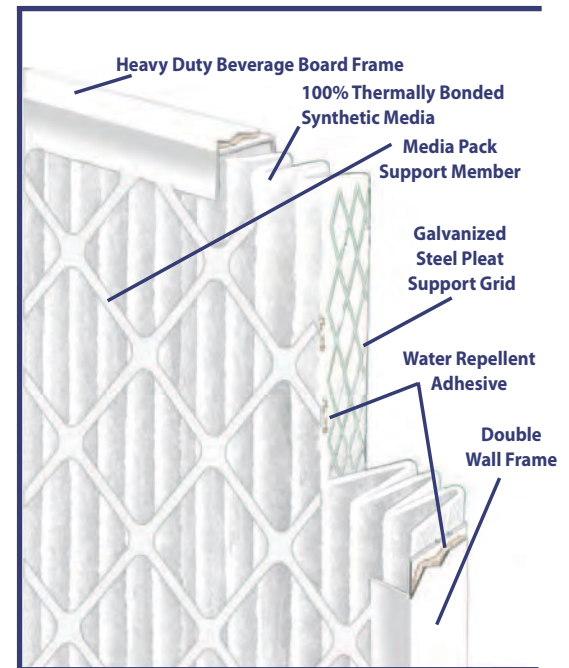
For over 40 years, Airguard® has been providing the high-performance DP® filter at competitive prices. Using ground-breaking R&D to provide a lower pressure drop and higher dust holding, the DP/DP Max filters are better than ever. And with the largest inventory of pleats in the industry, availability is never an issue. Contact an Airguard representative for a quote today.

Media:

- 100% thermally bonded synthetic, proprietary media resists moisture and damage
- Mechanical efficiency – achieves MERV 8 rating before and after a conditioning step
- Independently tested and verified for quality control
- Extremely low pressure drop and high dust holding capacity

Construction:

- Heavy duty, moisture resistant beverage board frame
- Two-piece frame provides rigidity with double-wall thickness around outer edge
- Die-cut pattern provides 50% more contact points between media pack and frame
- Integral die-cut cross members provide strength and rigidity
- 4" models use die-cut finger separators to stabilize each pleat
- Water repellent adhesive used to bond frame with media pack
- Rust-resistant, expanded metal support grid made of galvanized steel
- 100% water repellent, high-strength adhesive coats entire interior of frame
- Pleat tips of media pack bonded to diagonal support members



All DP filters are designed with a consistent pleat shape on predetermined centers causing dirt to collect evenly over the entire surface of the media. Fully utilizing every square inch of media results in a slow, steady rise in resistance for maximum dust holding capacity.

Pleat Consistency:

Airguard uses sophisticated production control techniques to ensure consistent pleat count, height, shape, and spacing. This uniform spacing contributes to a better-performing pleat with low pressure drop, full depth dust loading, and longer service life.

Durability:

Because of moisture resistant media and construction, the pleats hold together even when wet, preventing delaminating, excessive buckling, collapsing, racking, warping, and bending. Additionally, the galvanized steel metal grid resists rust, maintains pleat shape, and prevents fluttering in even the most rugged conditions.

Compare:

Compare the performance and value of DP pleats to other types of filters, including disposable panels, ring panels, pads and frames, and permanent filters. The DP line has significantly lower initial resistance levels to aid in your energy savings initiatives, and it also outperforms the alternatives in efficiency, dust holding capacity, durability, and price.

NOMINAL SIZE (2)	ACTUAL SIZE	AIRFLOW 300 FPM	AIRFLOW 500 FPM	AIRFLOW 625 FPM	INITIAL RESISTANCE DP	INITIAL RESISTANCE DP MAX	MEDIA AREA DP	MEDIA AREA DP MAX
10x10x1	9-1/2 x 9-1/2 x 3/4	210	350	N/R	0.14	0.16	1.6	1.5
10x20x1	9-1/2 x 19-1/2 x 3/4	415	700	N/R	0.14	0.16	2.6	2.4
10x24x1	9-3/8 x 23-3/8 x 3/4	500	830	N/R	0.14	0.16	3.1	2.9
12x12x1	11-3/4 x 11-3/4 x 3/4	300	500	N/R	0.14	0.16	1.9	1.8
12x16x1	11-1/2 x 15-3/4 x 3/4	400	670	N/R	0.14	0.16	2.5	2.3
12x20x1	11-1/2 x 19-1/2 x 3/4	500	830	N/R	0.14	0.16	3.1	2.9
12x24x1	11-3/8 x 23-3/8 x 3/4	600	1000	N/R	0.14	0.16	3.7	3.4
14x14x1	13-3/4 x 13-3/4 x 3/4	410	680	N/R	0.14	0.16	2.7	2.4
14x20x1	13-1/2 x 19-1/2 x 3/4	585	980	N/R	0.14	0.16	3.8	3.4
14x24x1	13-3/8 x 23-3/8 x 3/4	700	1170	N/R	0.14	0.16	4.6	4.0
14x25x1	13-1/2 x 24-1/2 x 3/4	730	1220	N/R	0.14	0.16	4.8	4.2
14x30x1*	13-3/4 x 29-3/4 x 3/4	875	1460	N/R	0.14	0.16	5.7	5.0
15x20x1	14-1/2 x 19-1/2 x 3/4	625	1050	N/R	0.14	0.16	4.0	3.6
15x30x1*	14-3/4 x 29-3/4 x 3/4	935	1560	N/R	0.14	0.16	6.1	5.4
16x16x1	15-1/2 x 15-1/2 x 3/4	530	890	N/R	0.14	0.16	4.0	3.1
16x20x1	15-1/2 x 19-1/2 x 3/4	665	1110	N/R	0.14	0.16	4.3	3.8
16x24x1	15-3/4 x 23-3/8 x 3/4	800	1330	N/R	0.14	0.16	5.1	4.6
16x25x1	15-1/2 x 24-1/2 x 3/4	835	1400	N/R	0.14	0.16	5.3	4.8
16x30x1*	15-3/4 x 29-3/4 x 3/4	1000	1670	N/R	0.14	0.16	6.5	5.8
18x18x1	17-3/4 x 17-3/4 x 3/4	675	1125	N/R	0.14	0.16	4.3	3.7
18x20x1	17-3/8 x 19-1/2 x 3/4	750	1250	N/R	0.14	0.16	4.8	4.0
18x22x1	17-3/8 x 21-1/2 x 3/4	825	1375	N/R	0.14	0.16	5.7	4.8
18x24x1	17-3/8 x 23-3/8 x 3/4	900	1500	N/R	0.14	0.16	5.7	4.8
18x25x1	17-1/2 x 24-1/2 x 3/4	935	1560	N/R	0.14	0.16	6.0	5.1
20x20x1	19-1/2 x 19-1/2 x 3/4	830	1400	N/R	0.14	0.16	5.5	4.8
20x22x1	19-3/4 x 21-3/4 x 3/4	915	1525	N/R	0.14	0.16	6.6	5.7
20x24x1	19-3/8 x 23-3/8 x 3/4	1000	1670	N/R	0.14	0.16	6.6	5.7
20x25x1	19-1/2 x 24-1/2 x 3/4	1040	1730	N/R	0.14	0.16	6.9	6.0
20x30x1*	19-1/2 x 29-1/2 x 3/4	1250	2170	N/R	0.14	0.16	8.1	7.2
22x22x1	21-3/4 x 21-3/4 x 3/4	1005	2080	N/R	0.14	0.16	7.1	5.0
24x24x1	23-3/8 x 23-3/8 x 3/4	1200	2000	N/R	0.14	0.16	7.7	6.6
24x30x1	23-3/4 x 29-3/4 x 3/4	1500	2500	N/R	0.14	0.16	9.7	8.5
25x25x1	24-1/2 x 24-1/2 x 3/4	1300	2170	N/R	0.14	0.16	8.7	7.5
10x20x2	9-1/2 x 19-1/2 x 1-3/4	415	700	870	0.19	0.21	6.2	4.1
12x12x2	11-3/4 x 11-3/4 x 1-3/4	300	500	625	0.19	0.21	4.4	3.1
12x20x2	11-1/2 x 19-1/2 x 1-3/4	500	840	1040	0.19	0.21	7.2	5.2
12x24x2	11-3/8 x 23-3/8 x 1-3/4	600	1000	1250	0.19	0.21	8.6	6.2
14x20x2	13-1/2 x 19-1/2 x 1-3/4	585	980	1220	0.19	0.21	8.8	5.7
14x25x2	13-1/2 x 24-1/2 x 1-3/4	730	1220	1520	0.19	0.21	11.0	7.1
15x20x2	14-1/2 x 19-1/2 x 1-3/4	625	1050	1300	0.19	0.21	9.3	6.2
16x16x2	15-3/4 x 15-3/4 x 1-3/4	530	890	1110	0.19	0.21	9.8	6.7
16x20x2	15-1/2 x 19-1/2 x 1-3/4	665	1120	1400	0.19	0.21	9.8	6.7
16x24x2	15-3/8 x 23-3/8 x 1-3/4	800	1340	1670	0.19	0.21	11.7	8.0
16x25x2	15-1/2 x 24-1/2 x 1-3/4	835	1400	1740	0.19	0.21	11.7	8.0
18x18x2	17-3/4 x 17-3/4 x 1-3/4	675	1125	1410	0.19	0.21	10.3	7.0
18x20x2	17-1/2 x 19-1/2 x 1-3/4	750	1250	1560	0.19	0.21	11.3	7.7
18x22x2	17-1/2 x 21-1/2 x 1-3/4	825	1375	1720	0.19	0.21	13.6	9.3
18x24x2	17-3/8 x 23-3/8 x 1-3/4	900	1500	1875	0.19	0.21	13.6	9.3
18x25x2	17-1/2 x 24-1/2 x 1-3/4	935	1570	1950	0.19	0.21	14.2	9.7
20x20x2	19-1/2 x 19-1/2 x 1-3/4	830	1400	1740	0.19	0.21	12.4	8.3
20x24x2	19-3/8 x 23-3/8 x 1-3/4	1000	1670	2080	0.19	0.21	14.8	9.9
20x25x2	19-1/2 x 24-1/2 x 1-3/4	1040	1750	2170	0.19	0.21	15.5	10.3
20x30x2*	19-1/2 x 29-1/2 x 1-3/4	1250	2085	2600	0.19	0.21	18.6	12.9
24x24x2	23-3/8 x 23-3/8 x 1-3/4	1200	2000	2500	0.19	0.21	17.9	11.7
25x25x2	24-1/2 x 24-1/2 x 1-3/4	1300	2170	2710	0.19	0.21	20.0	12.9
12x24x4	11 3/8 x 23 3/8 x 3-3/4	600	1000	1250	0.17	0.19	12.4	11.2
16x20x4	15 1/2 x 19 1/2 x 3-3/4	665	1120	1400	0.17	0.19	14.5	12.5
16x25x4	15 1/2 x 24 1/2 x 3-3/4	835	1400	1740	0.17	0.19	18.3	15.6
18x24x4	17 3/8 x 23 3/8 x 3-3/4	900	1500	1875	0.17	0.19	19.9	16.2
20x20x4	19 1/2 x 19 1/2 x 3-3/4	830	1400	1740	0.17	0.19	18.7	15.6
20x24x4	19 3/8 x 23 3/8 x 3-3/4	1000	1670	2080	0.17	0.19	22.4	18.7
20x25x4	19 1/2 x 24 1/2 x 3-3/4	1040	1750	2170	0.17	0.19	23.5	19.6
24x24x4	23 3/8 x 23 3/8 x 3-3/4	1200	2000	2500	0.17	0.19	27.4	22.4
25x29x4*	24-1/2 x 28-1/2 x 3-3/4	1280	2430	2660	0.17	0.19	32.6	26.1
28x30x4*	27-1/2 x 29-1/2 x 3-3/4	1750	2920	3640	0.17	0.19	38.5	31.1
12x24x6	11 3/8 x 23 3/8 x 5 7/8	600	1000	1250	-	0.18	-	16.9
16x20x6	15 1/2 x 19 1/2 x 5 7/8	665	1120	1400	-	0.18	-	18.9
16x25x6	15 1/2 x 24 1/2 x 5 7/8	835	1400	1740	-	0.18	-	23.6
18x24x6	17 3/8 x 23 3/8 x 5 7/8	900	1500	1875	-	0.18	-	26.3
20x20x6	19 1/2 x 19 1/2 x 5 7/8	830	1400	1740	-	0.18	-	23.7
20x24x6	19 3/8 x 23 3/8 x 5 7/8	1000	1670	2080	-	0.18	-	28.2
20x25x6	19 1/2 x 24 1/2 x 5 7/8	1040	1750	2170	-	0.18	-	29.5
24x24x6	23 3/8 x 23 3/8 x 5 7/8	1200	2000	2500	-	0.18	-	33.8

* Reverse Pleat

1. DP/ DP Max are MERV 8-A, tested per ASHRAE standard 52.2-2012 Appendix J. Test data based on 295 FPM for 24x24x1 and 492 FPM for 24x24x2 and 24x24x4 nominal sizes.
2. Filters may be installed with the pleats either vertical (preferred) or horizontal.
3. 6" deep sizes are available in DP Max models only.

DP®/DP Max

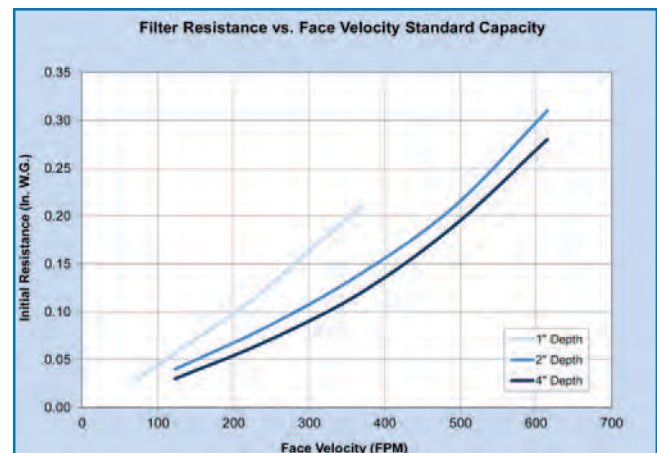
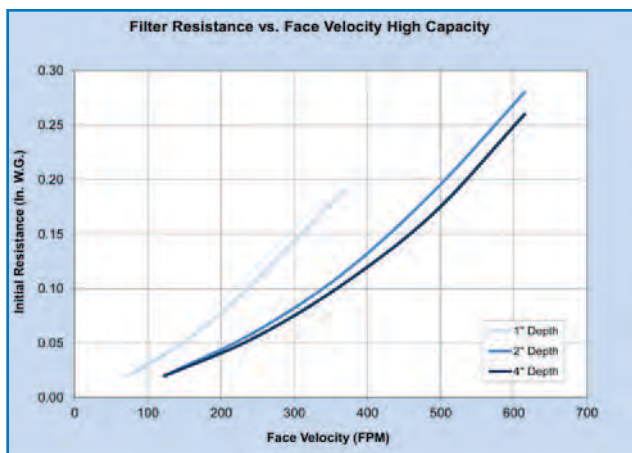
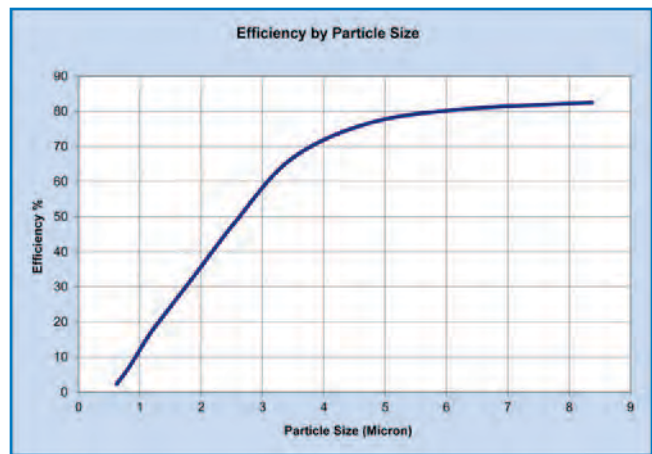
Extended Surface Pleated Filters

Value-Added Summary:

- Synthetic media does not support microbial growth
- Low pressure drop with high dust holding
- Mechanical efficiency
- Durable and damage resistant construction
- Engineered and designed for high strength and rigidity
- Pleat consistency maximizes performance
- Water and moisture resistant media, frame, and adhesive
- Proudly made in the USA

Applications:

- Virtually any application or air handling system
- Built-up filter banks, central station air handlers, rooftop units, side access housings
- Final filters for MERV 8 requirements
- Pre-filters in multi-stage systems



Pleat Count -	1"	2"	4"	6"
	(Pleats per foot)			
DP -	14.0	15.0	11.0	-
DP Max -	12.0	10.0	9.0	9.0

Underwriters Laboratories, Inc. Classification:

DP and DP Max filters are classified per UL Standard 900 for flammability.

Operating Temperature Limit: Maximum operating temperature is 225°F (107°C).

A-DP-0416



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CLARCOR Air Filtration Products
 100 River Ridge Circle • Jeffersonville, IN 47130
 Customer Service: 1-866-247-4827 • Fax: 1-866-601-1809
 Email: mailbag@airguard.com • www.clair.com/airguard

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PD:2025.02.06

ENERGY RECOVERY UNIT (DX & GAS) SCHEDULE

TAG	AREA SERVED	BASIS OF DESIGN		TOTAL CFM	SUPPLY FAN					EXHAUST FAN					COOLING COIL				HOT GAS REHEAT				INCORRECT HEATING COIL										COMPRESSORS				WHEEL WINTER		WHEEL SUMMER		ELECTRICAL			WEIGHT (LBS.)	NOTES
		MANUFACTURER	MODEL		CFM	MOTOR QTY.	HP (E.A.)	BHP	ESP (K.W.G.)	RPM	CFM	MOTOR QTY.	HP (E.A.)	BHP	ESP	RPM	TOTAL (MMH)	SENS. (MMH)	EAT (DHWHS)	LAT (DHWHS)	CAPACITY (MMH)	LAT (DR)	FUEL TYPE	INPUT (MMH)	OUTPUT (MMH)	ESP. (PSI)	EAT (DR)	LAT (DR)	STAGES	CONV. SIZE	NO.	RLA	URA	TYPE	EAT (DHWHS)	LAT (DHWHS)	EAT (DHWHS)	LAT (DHWHS)	VOLTS/PH/Hz	MCA	MSCP				
ERH-1	BASEMENT	GREENHECK	RVE-45-S2D-20K-A1	5,375	5,375	1	7.5	5.32	1.50	1,717	4,800	1	5.0	3.04	1.25	1,418	286.6	172.5	80.1/97.9	50.9/50.9	191.3	83.9	NAT. GAS	300.0	213.0	81.0	43.7	85.5	4:1	3/4"	2	32.3/34.0	N/A/240	INVERTER SCROLL	5.5/1.7	43.7/37.0	88.9/75.0	80.1/68.0	208/3/60	123.4	150	4,577	1,2,3,4		
ERH-2	BASEMENT	GREENHECK	RVE-45-S2D-20K-A1	5,375	5,375	1	7.5	5.30	1.50	1,717	4,800	1	5.0	3.04	1.25	1,418	286.6	172.5	80.1/97.9	50.9/50.9	191.3	83.9	NAT. GAS	300.0	213.0	81.0	43.7	85.5	4:1	3/4"	2	32.3/34.0	N/A/240	INVERTER SCROLL	5.5/1.7	43.7/37.0	88.9/75.0	80.1/68.0	208/3/60	123.4	150	4,577	1,2,3,4		
ERH-3	1 ST / 2 ND FLOORS	GREENHECK	RVE-45-S2D-20K-A1	5,370	5,370	1	10.0	7.40	1.00	1,770	5,775	1	5.0	4.20	1.20	1,760	298.6	185.4	80.5/88.4	53.9/53.8	213.2	84.9	NAT. GAS	300.0	213.0	81.0	41.7	77.0	4:1	3/4"	2	32.3/34.0	N/A/240	INVERTER SCROLL	5.5/1.7	42.9/36.4	89.9/75.0	80.5/68.1	208/3/60	129.0	150	4,528	1,2,3,4		

- NOTES**
- PROVIDE COMPLETE PACKAGED UNIT WITH ALL CONTROLS AND ACCESSORIES FOR PROPER OPERATION, INCLUDING 1" P ROOF CURB, BUILDING STATIC PRESSURE SENSOR, SPACE HUMIDITY SENSOR, OUTSIDE AIR TEMPERATURE SENSOR, SUPPLY AIR TEMPERATURE SENSOR, SPACE TEMPERATURE SENSOR, WIRING ACCESS PANELS, VFD RATED DIRECT DRIVE MOTORS, CO2 SENSOR AND SUPPLY FAN CONTROL, DDC ACTUATOR, ENERGY WHEEL ECONOMIZER CONTROL, FROST CONTROL MODULATING WHEEL, LOW LEAKAGE DAMPERS, MERV-13 FILTRATION, REMOTE DISPLAY (150FT CORD), AND POWERED GR CONVENIENCE OUTLET.
 - PROVIDE UL OR ETL LISTED WITH CONTROL PANEL AND FUSED DISCONNECT FOR SINGLE POINT POWER CONNECTION.
 - INSTALL SUPPLY AND RETURN AIR SMOKE DETECTION FOR ALL UNITS WITH GREATER THAN 2000 CFM IN ACCORDANCE WITH NFPA 72 AND MANUFACTURERS INSTRUCTIONS. SMOKE DETECTOR SHALL BE FURNISHED AND WIRING BY E.C. (ELECTRICAL CONTRACTOR)
 - PROVIDE UNIT COLOR AS SELECTED BY ARCHITECT FROM MANUFACTURERS STANDARD COLORS.